

The Lifecycle of Mathlib PRs

Queueboard data

Bryan Chen, MI Retreat: May 22, 2026

Queueboard backend

- Q4 2025: migrated from git-based backend (35k+ JSON files in a git repo) to SQL database
- Q1 2026: zulip bot (notifications for auto-assignment, stale PRs), consume GitHub webhooks
- Q2 2026: close-pr command, import data from old queueboard, sync more data fields (comment / review comment metadata)

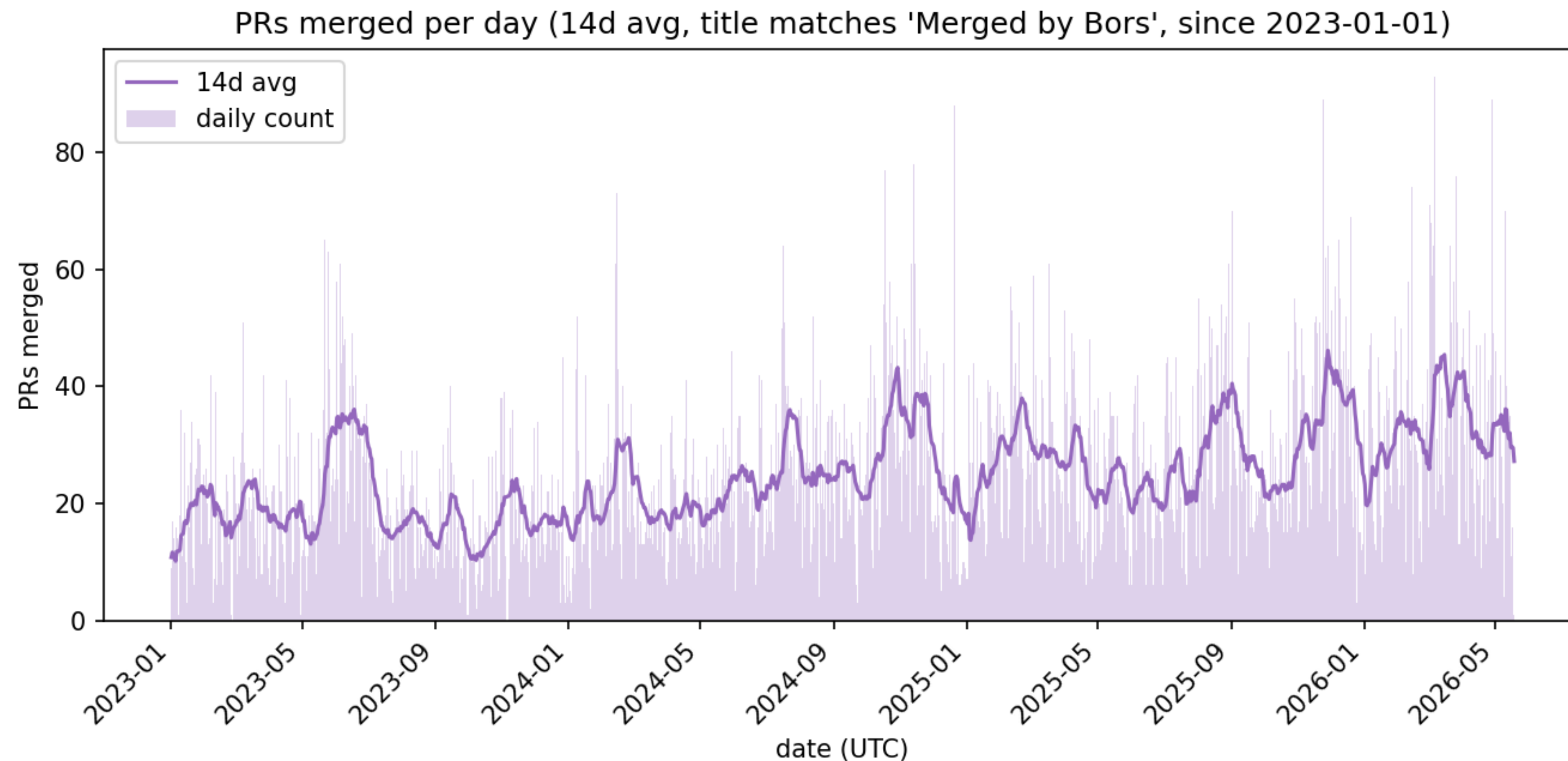
<https://github.com/leanprover-community/queueboard-core>

Queueboard backend data

- PRs (author, title, PR body, files touched, ...)
- Events (PR opened / closed, labeled, assigned, commented, reviewed)
 - Inline comment metadata (timestamp, commenter, which review event)
- Queue windows (intervals when a PR was “on the queue”)
- CI statuses (name, git SHA, timestamp, success/failure)

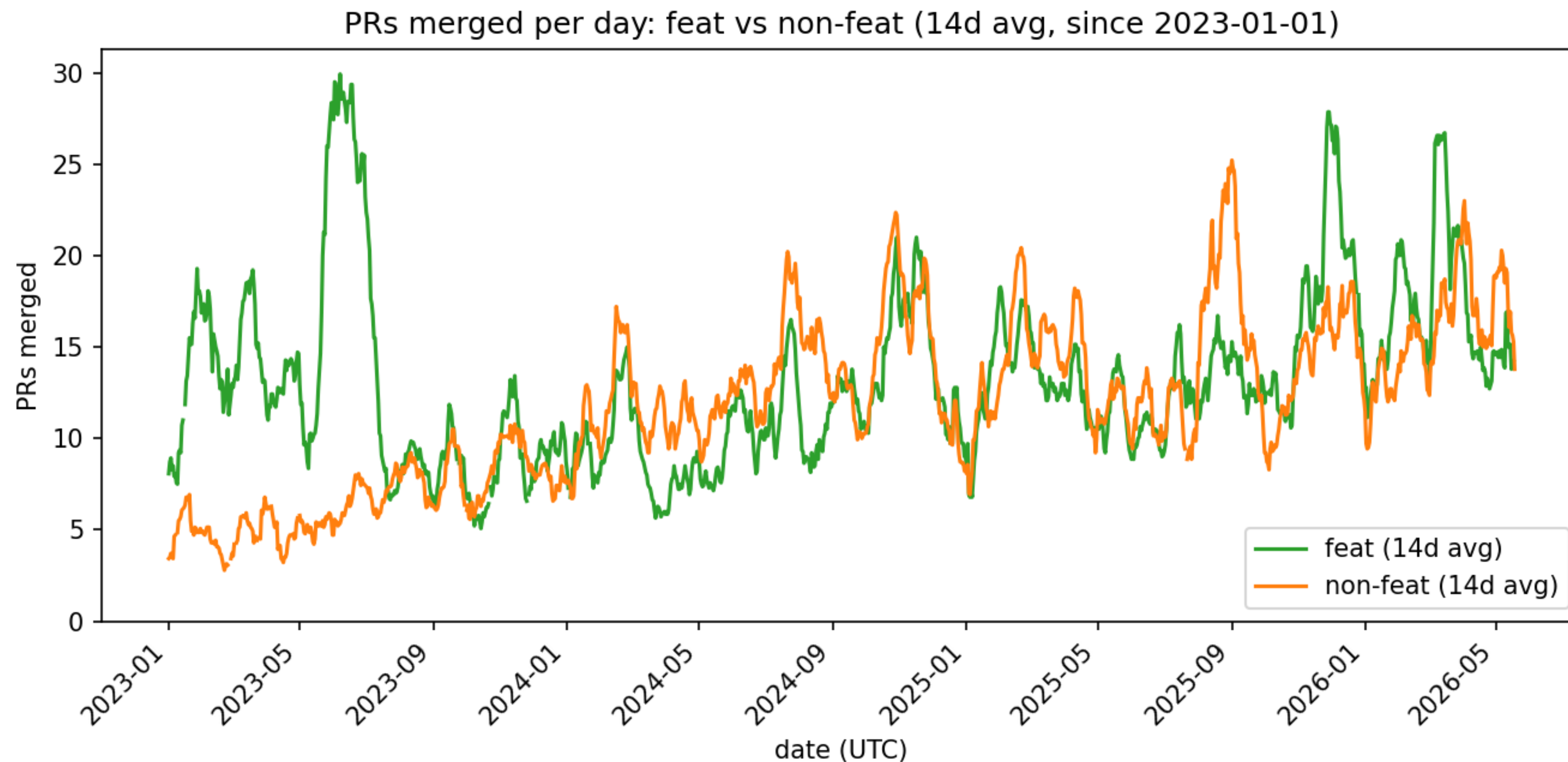
State of review

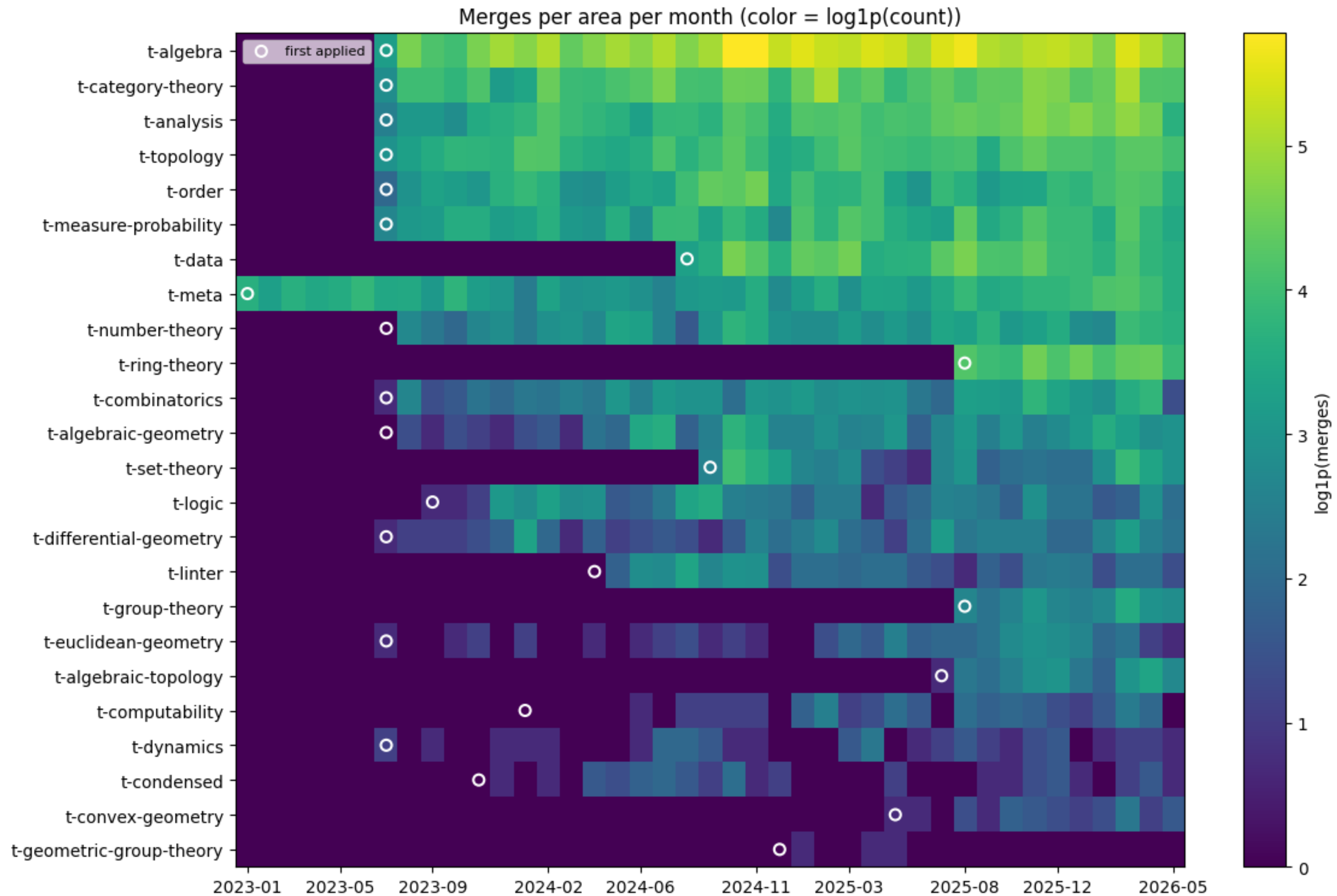
<https://mathlib-initiative.github.io/qb-notebook/>



State of review

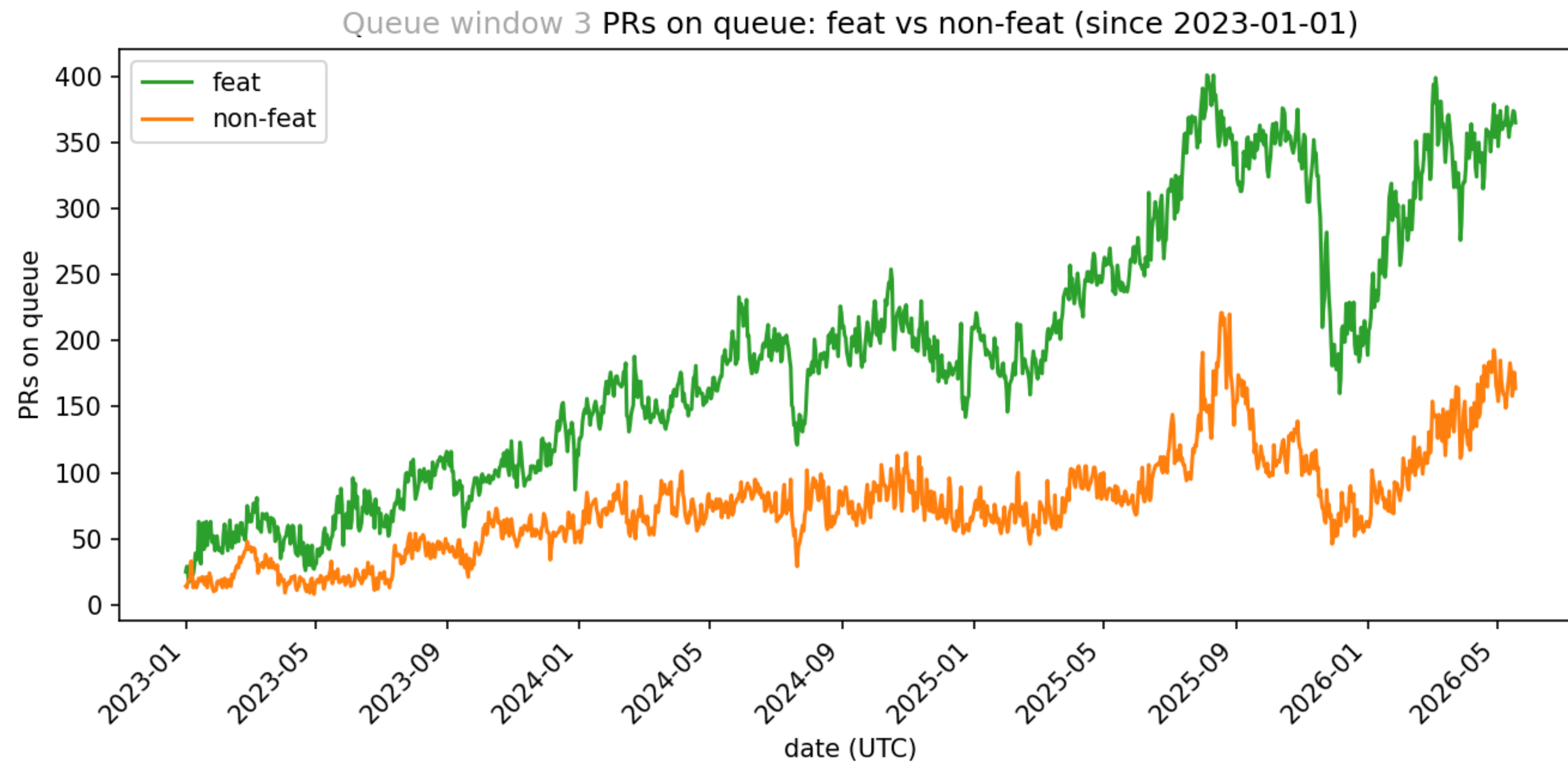
<https://mathlib-initiative.github.io/qb-notebook/>

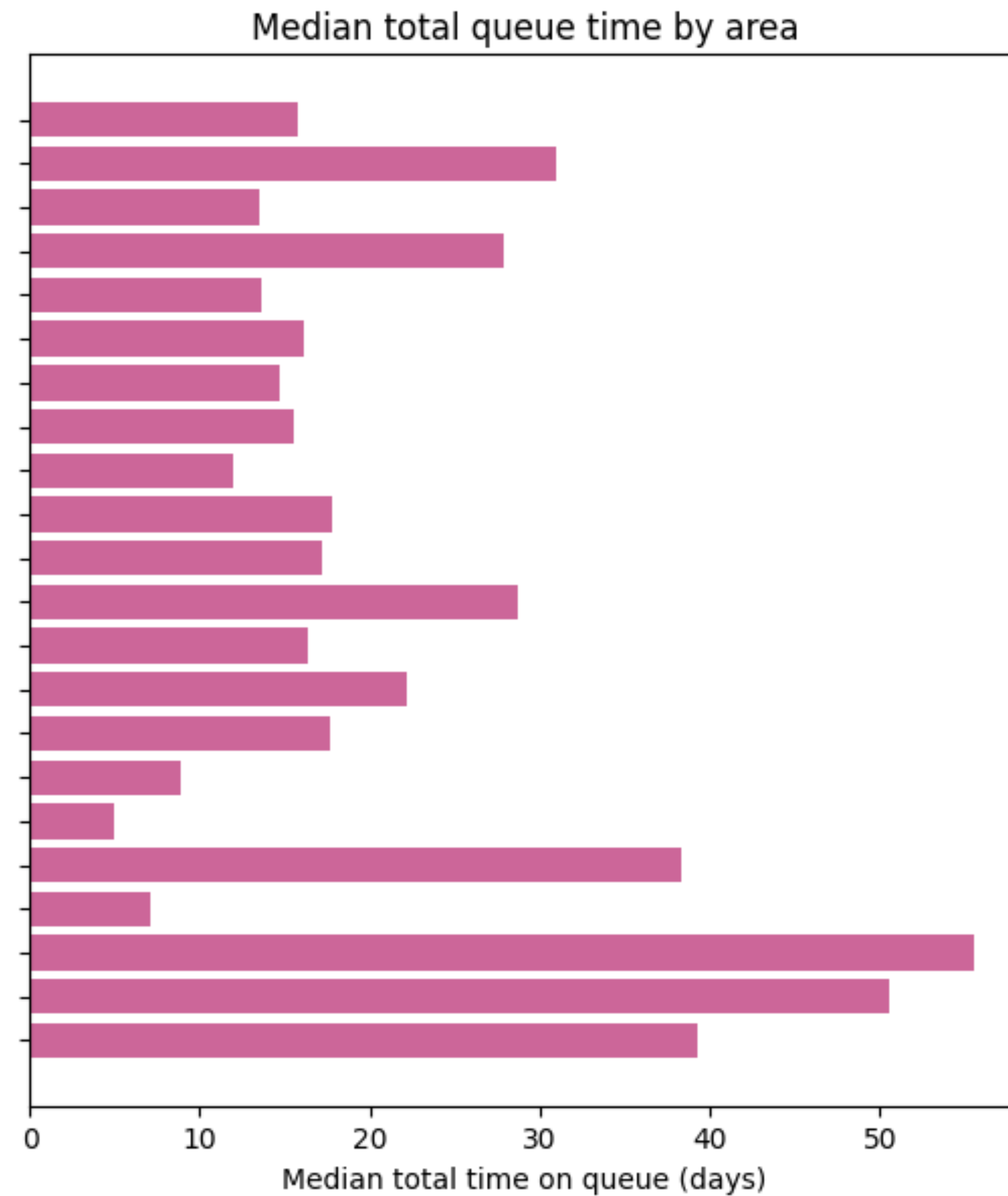
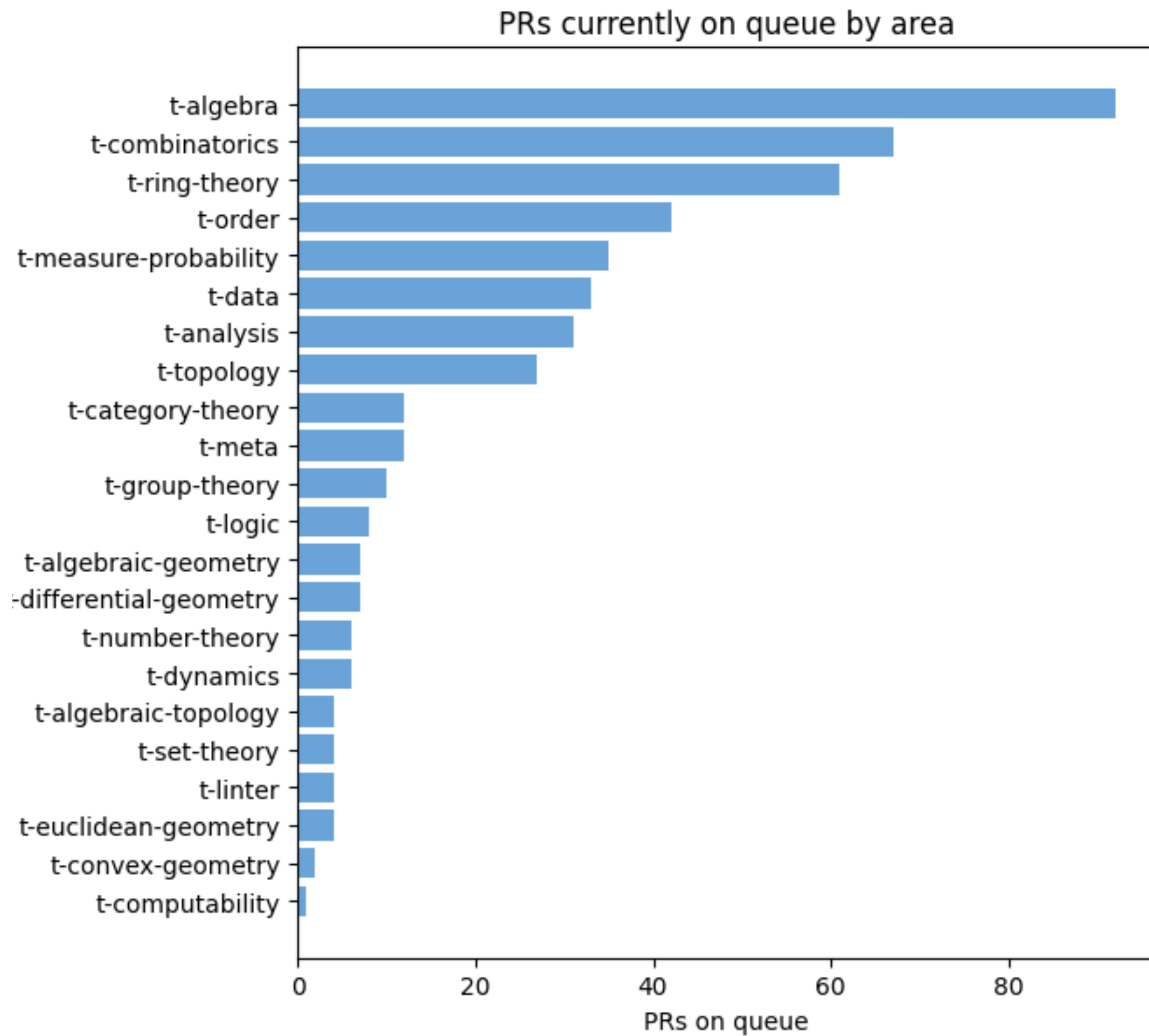




State of review

<https://mathlib-initiative.github.io/qb-notebook/>

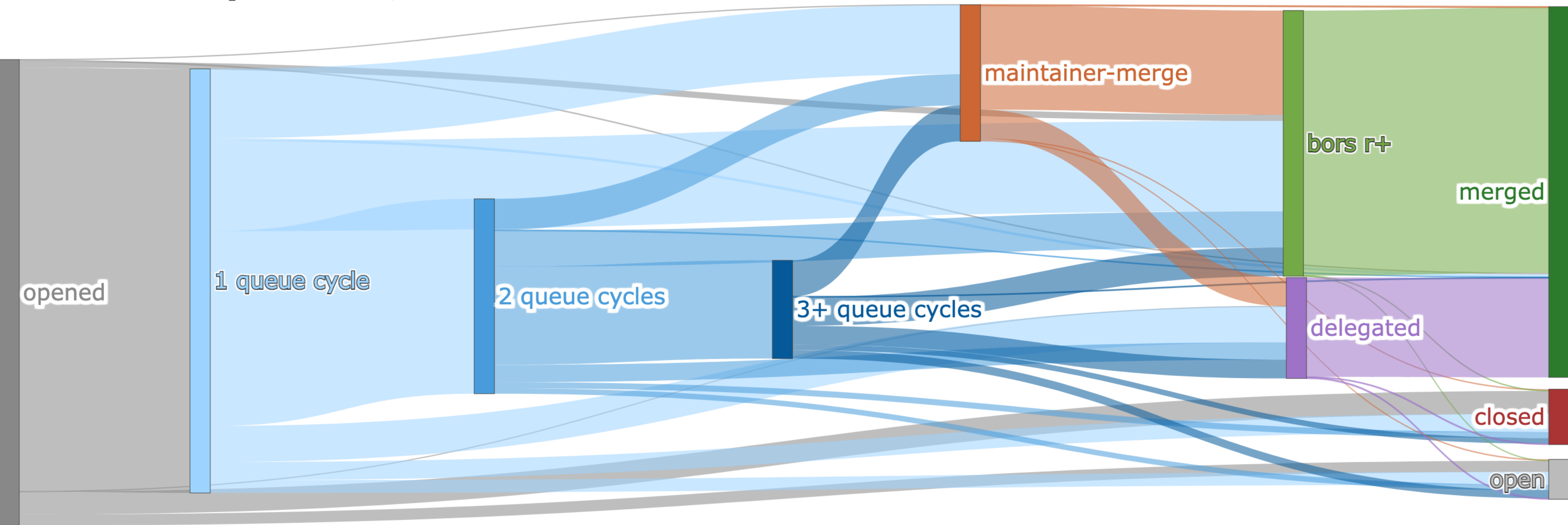




How do PRs progress through their life cycle?

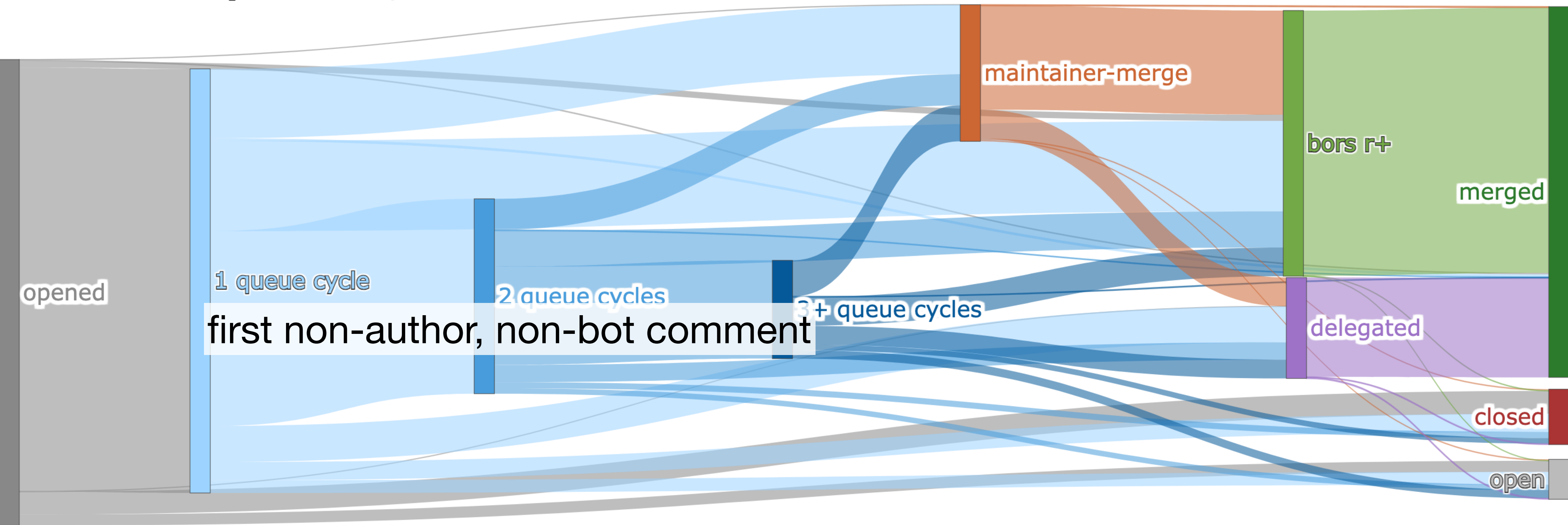
PRs opened between 2024-02-15 (maintainer-merge introduced) and 2026-05-21

Full lifecycle — 28,185 PRs



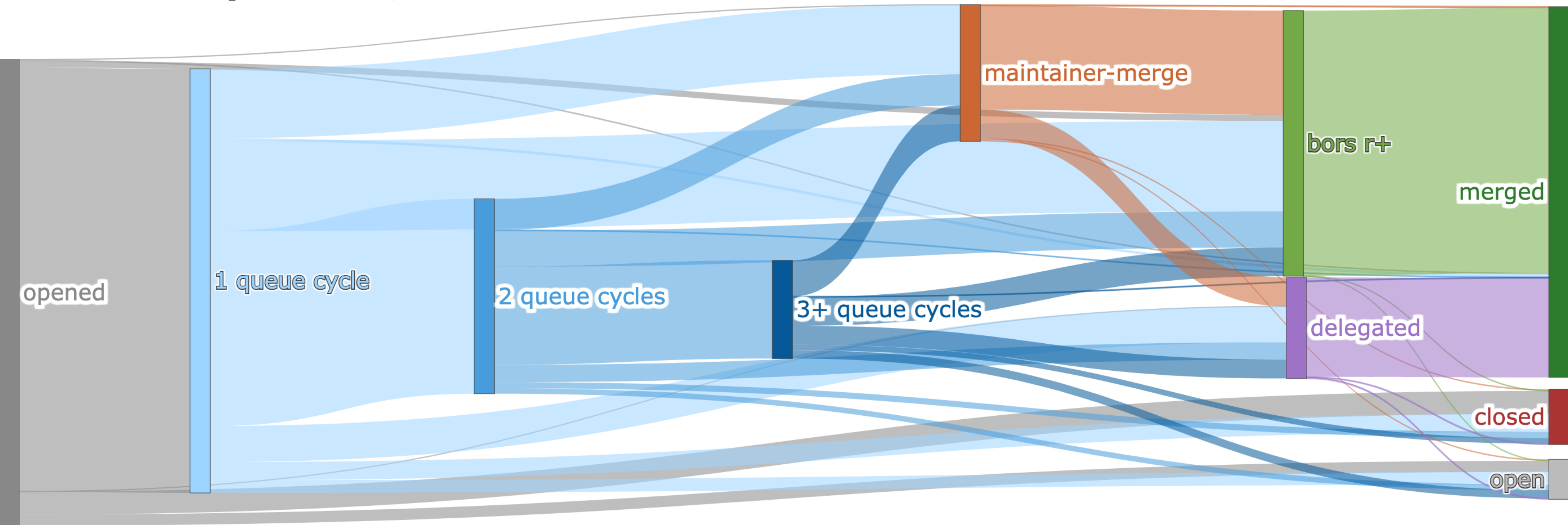
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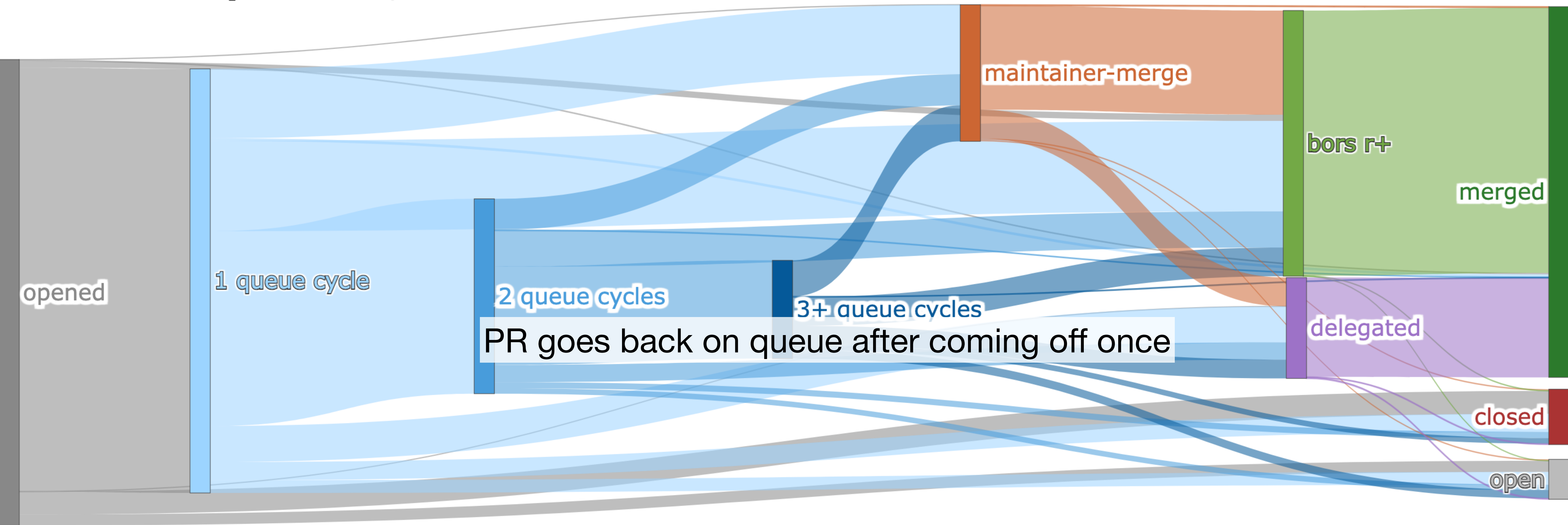
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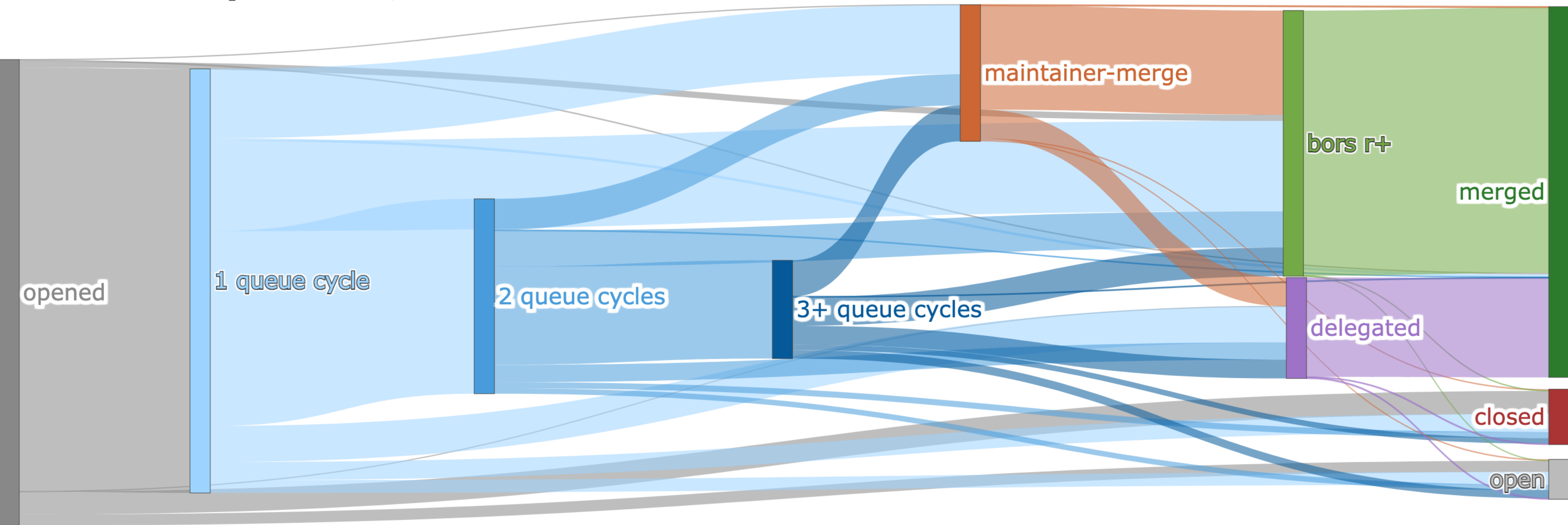
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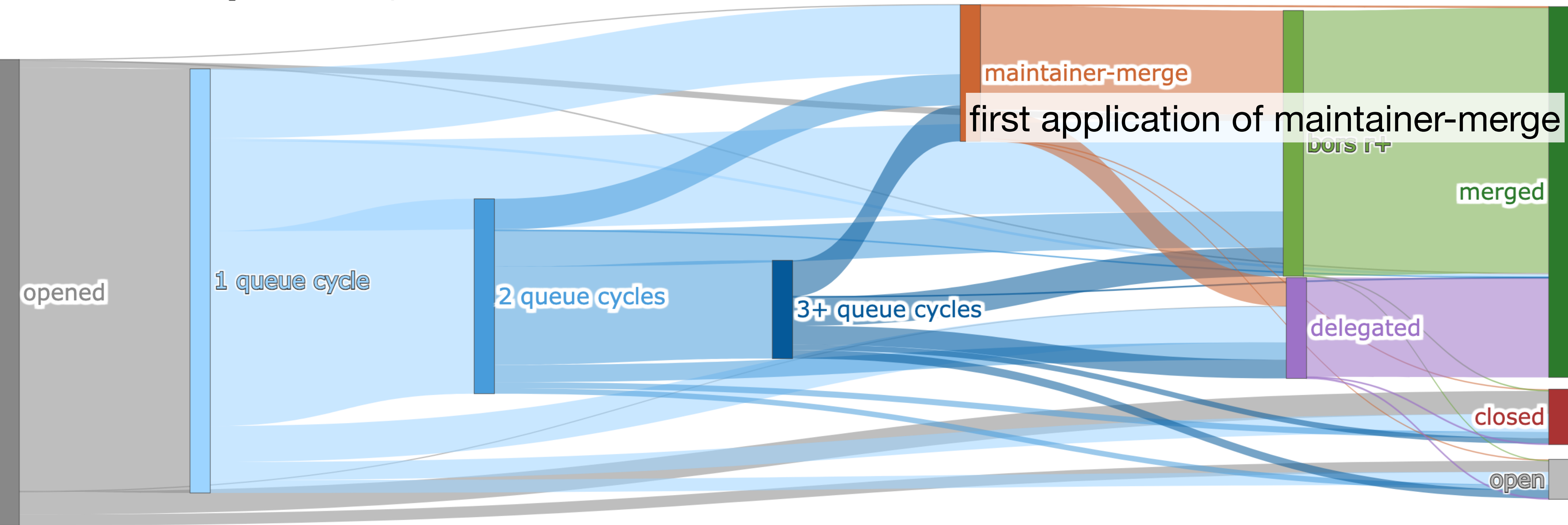
PRs opened between 2024-02-15 (maintainer-merge introduced) and 2026-05-21

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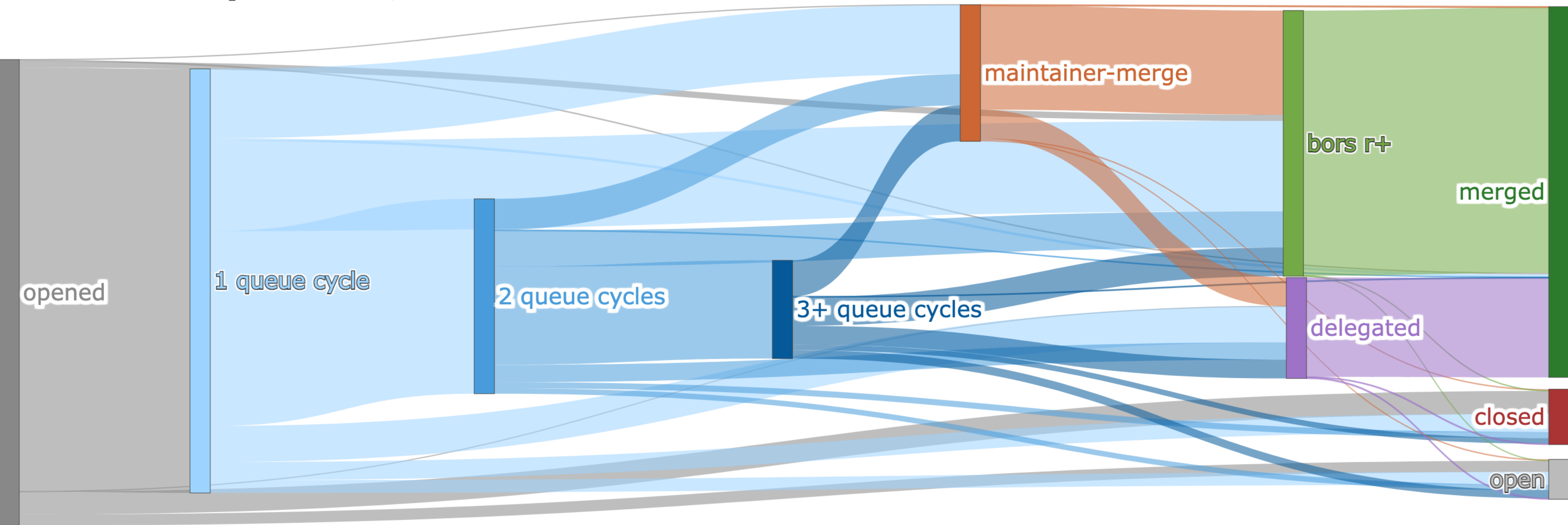
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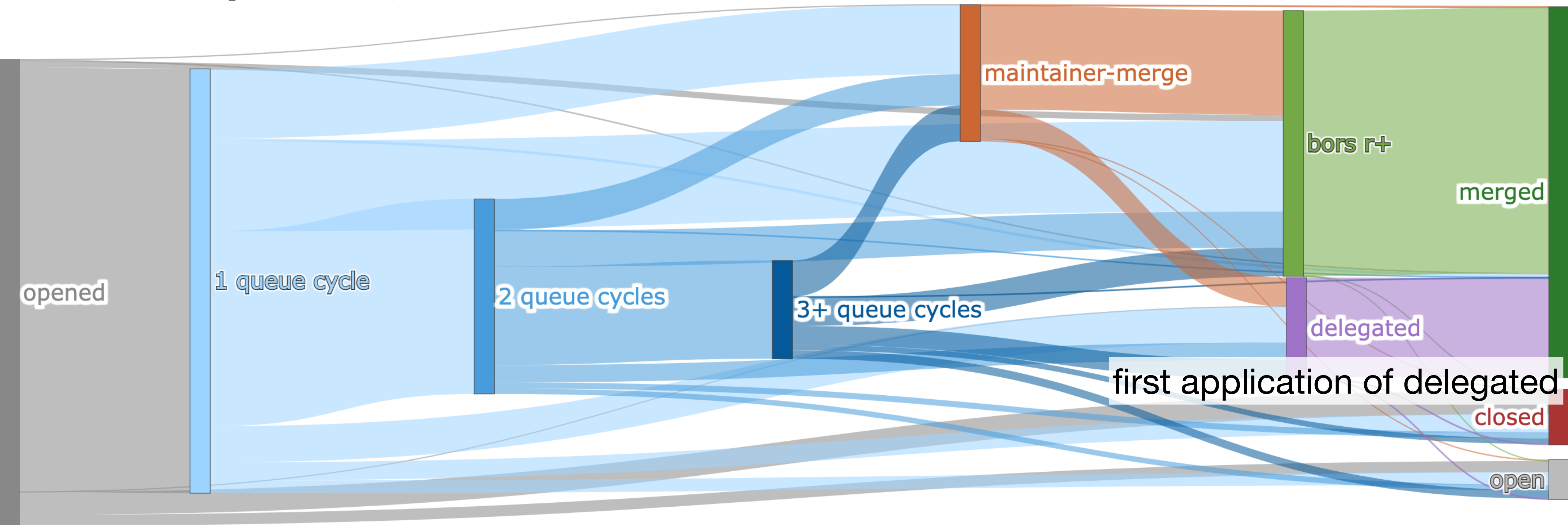
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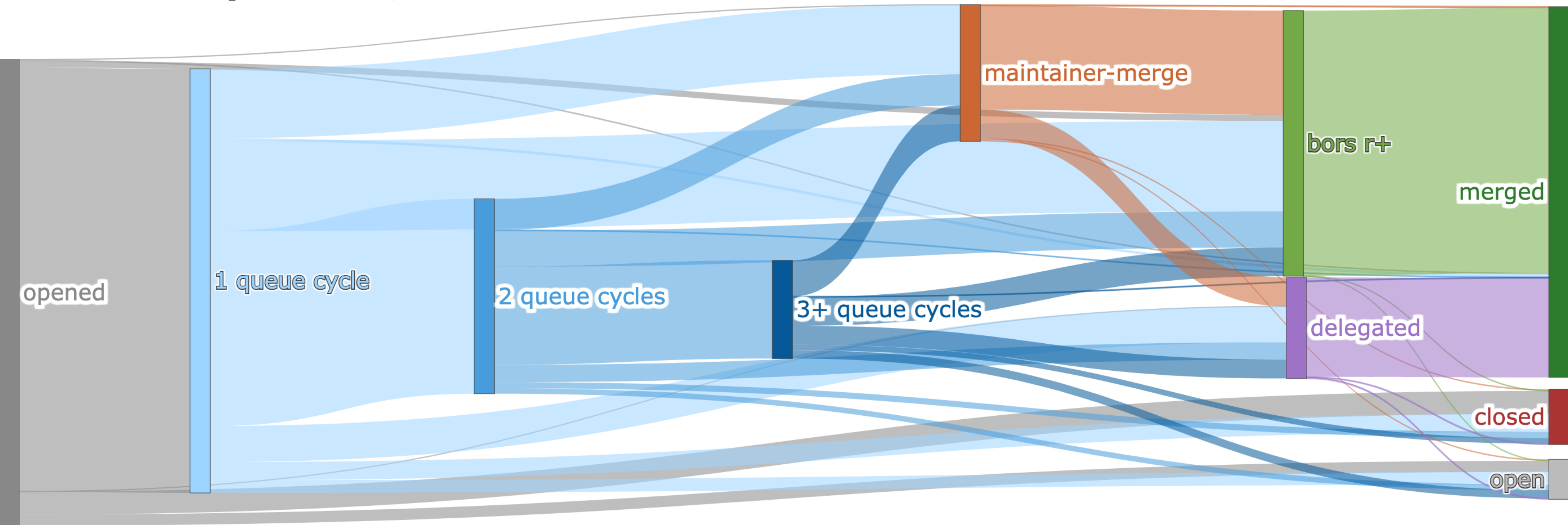
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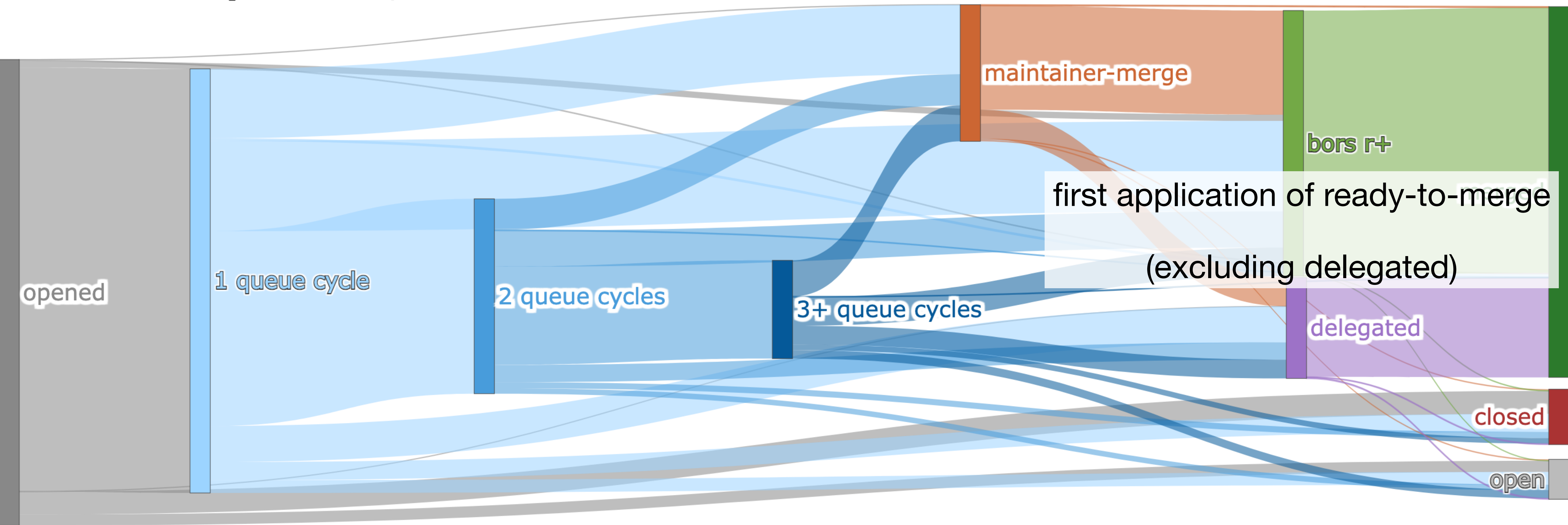
PRs opened between 2024-02-15 (maintainer-merge introduced) and 2026-05-21

Full lifecycle — 28,185 PRs



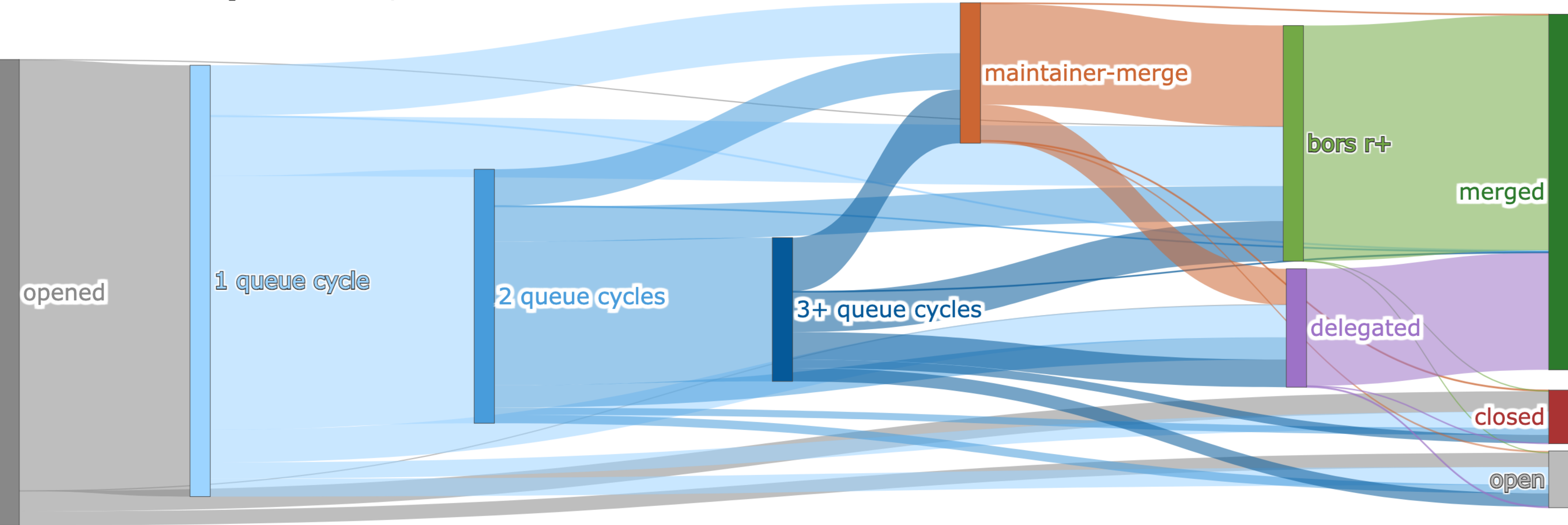
PRs opened between 2024-02-15 (maintainer-merge introduced) and 2026-05-21

Full lifecycle — 28,185 PRs



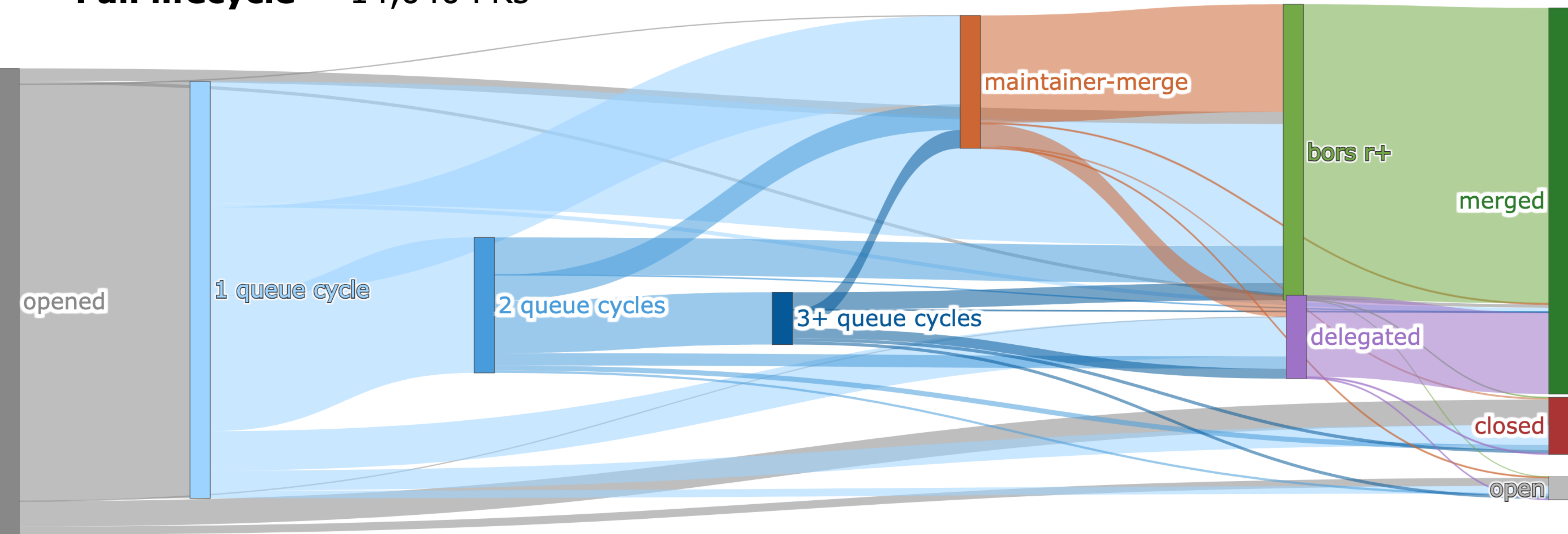
feat PRs opened between 2024-02-15 and 2026-05-21

Full lifecycle — 14,145 PRs

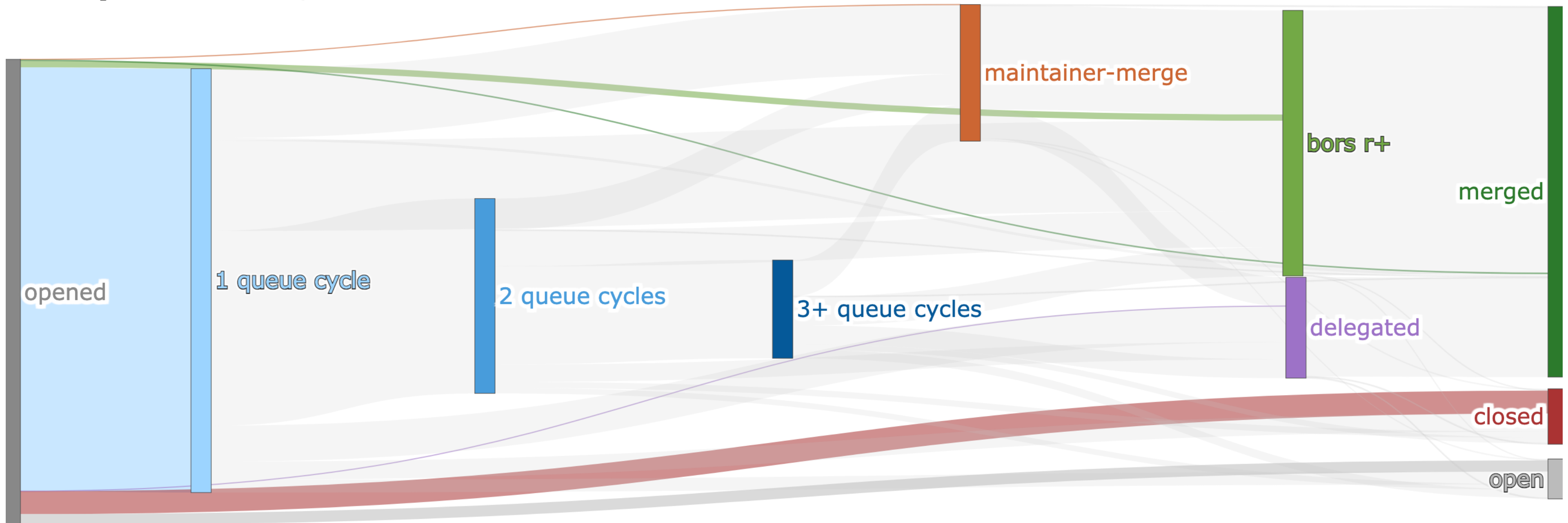


non-feat PRs opened between 2024-02-15 and 2026-05-21

Full lifecycle — 14,040 PRs



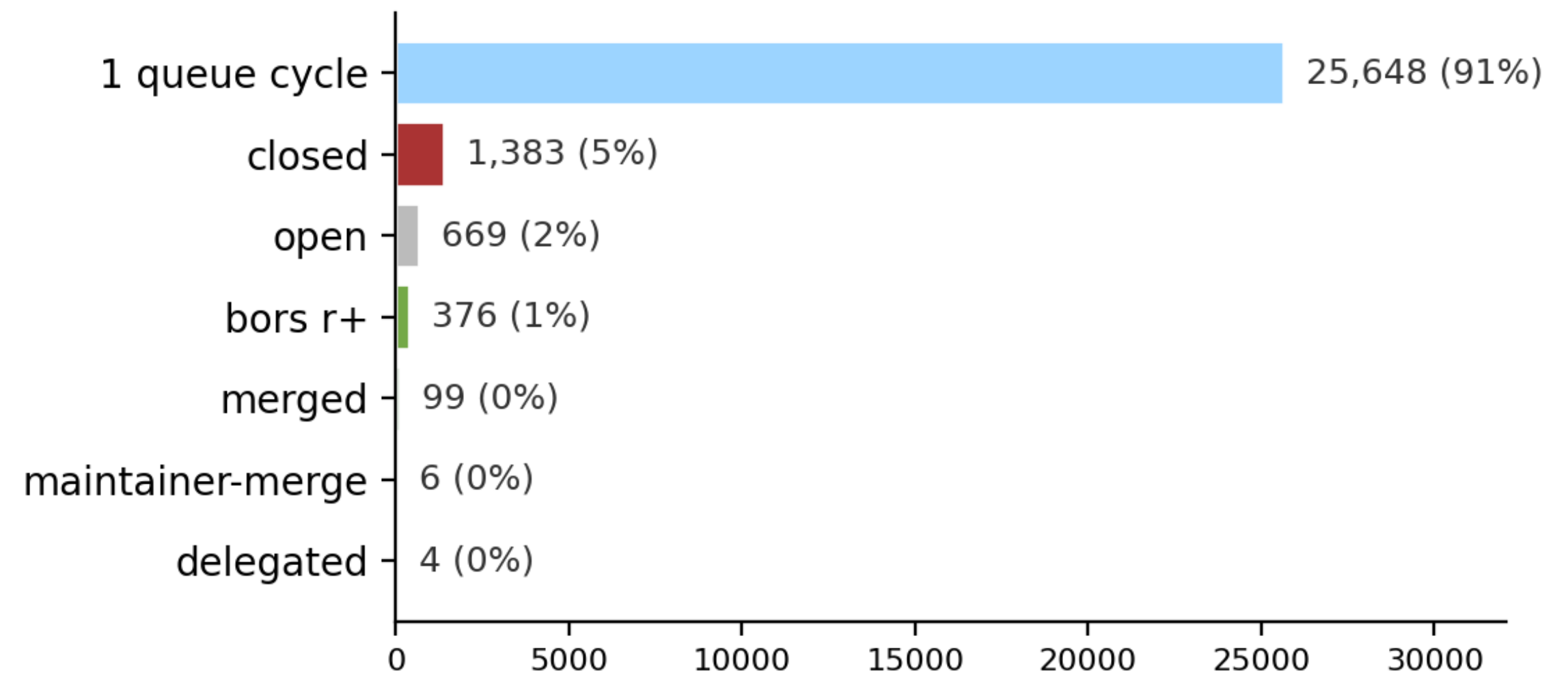
opened — 28,185 PRs



opened — flow breakdown

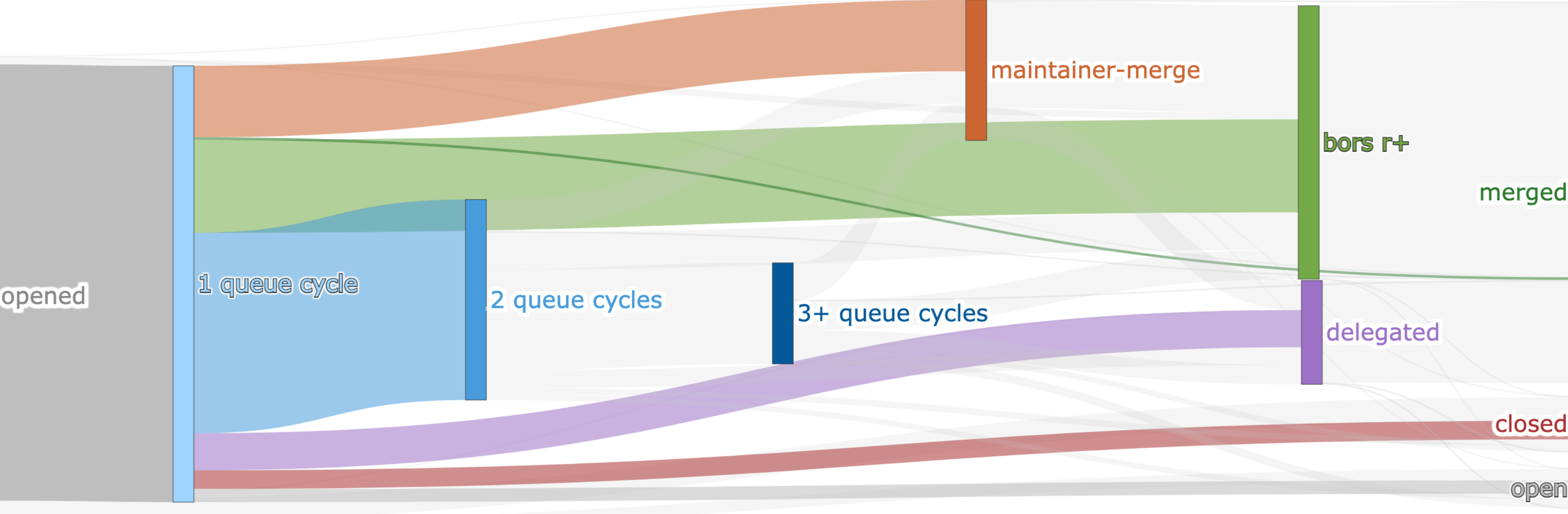
incoming — 0 PRs

outgoing — 28,185 PRs



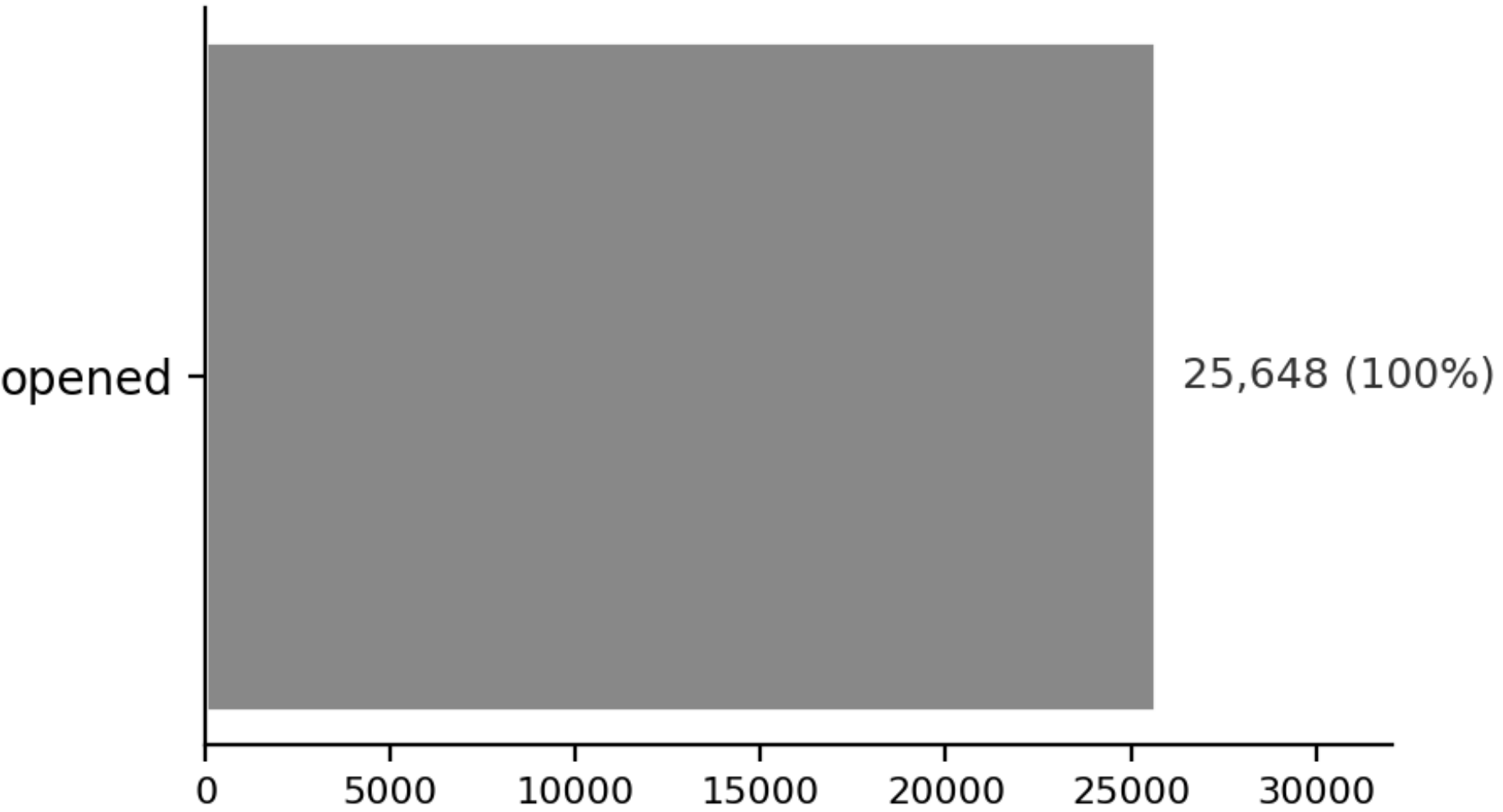
(first non-author, non-bot comment)

1 queue cycle — 25,648 PRs

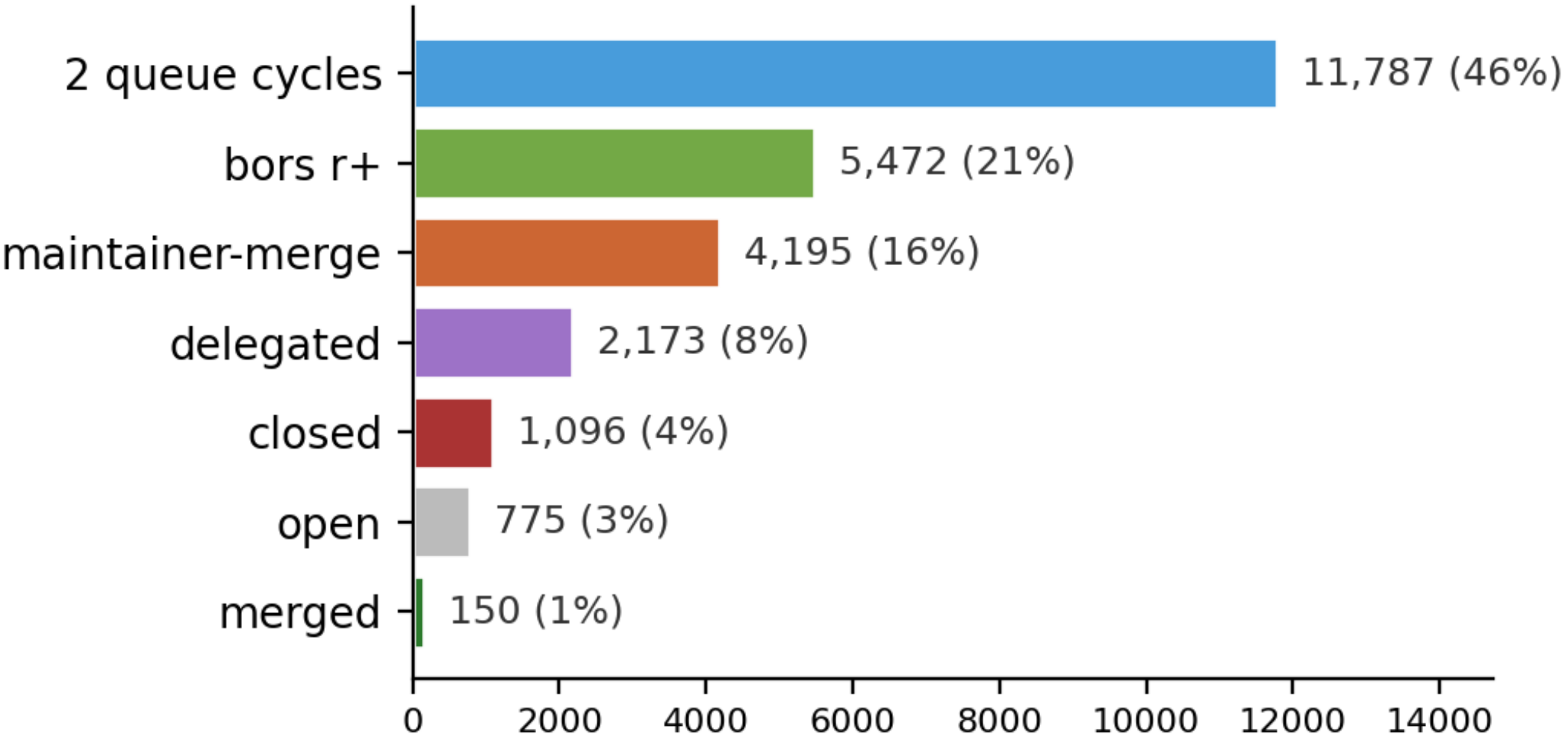


1 queue cycle — flow breakdown

incoming — 25,648 PRs

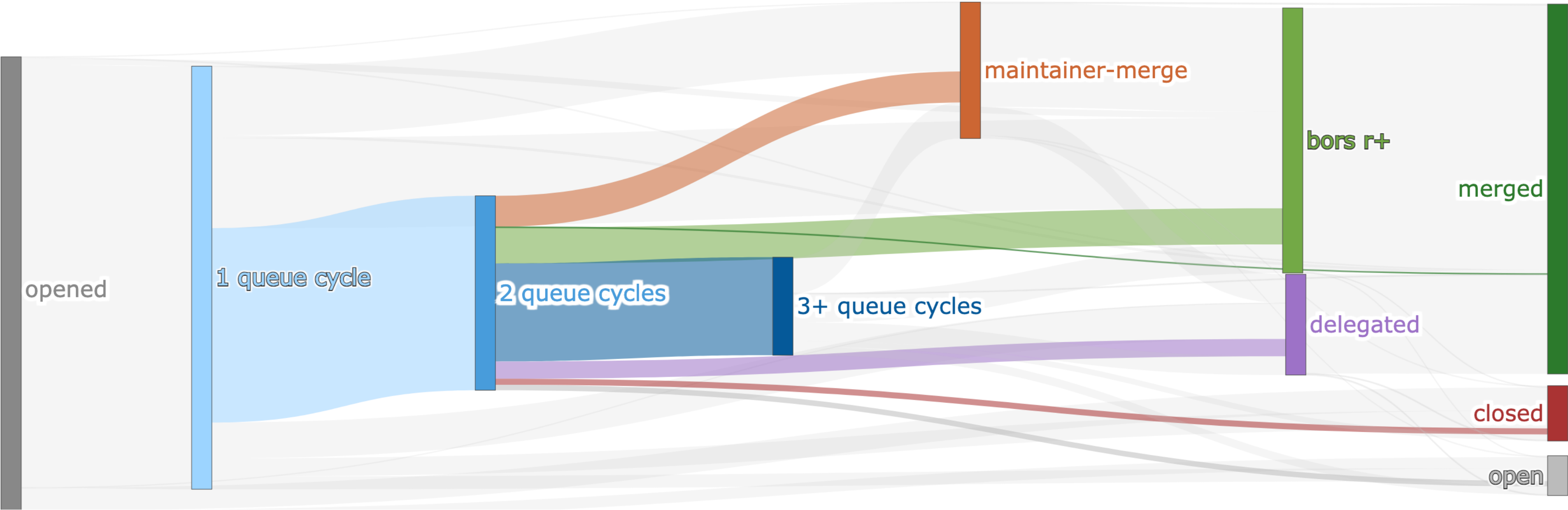


outgoing — 25,648 PRs



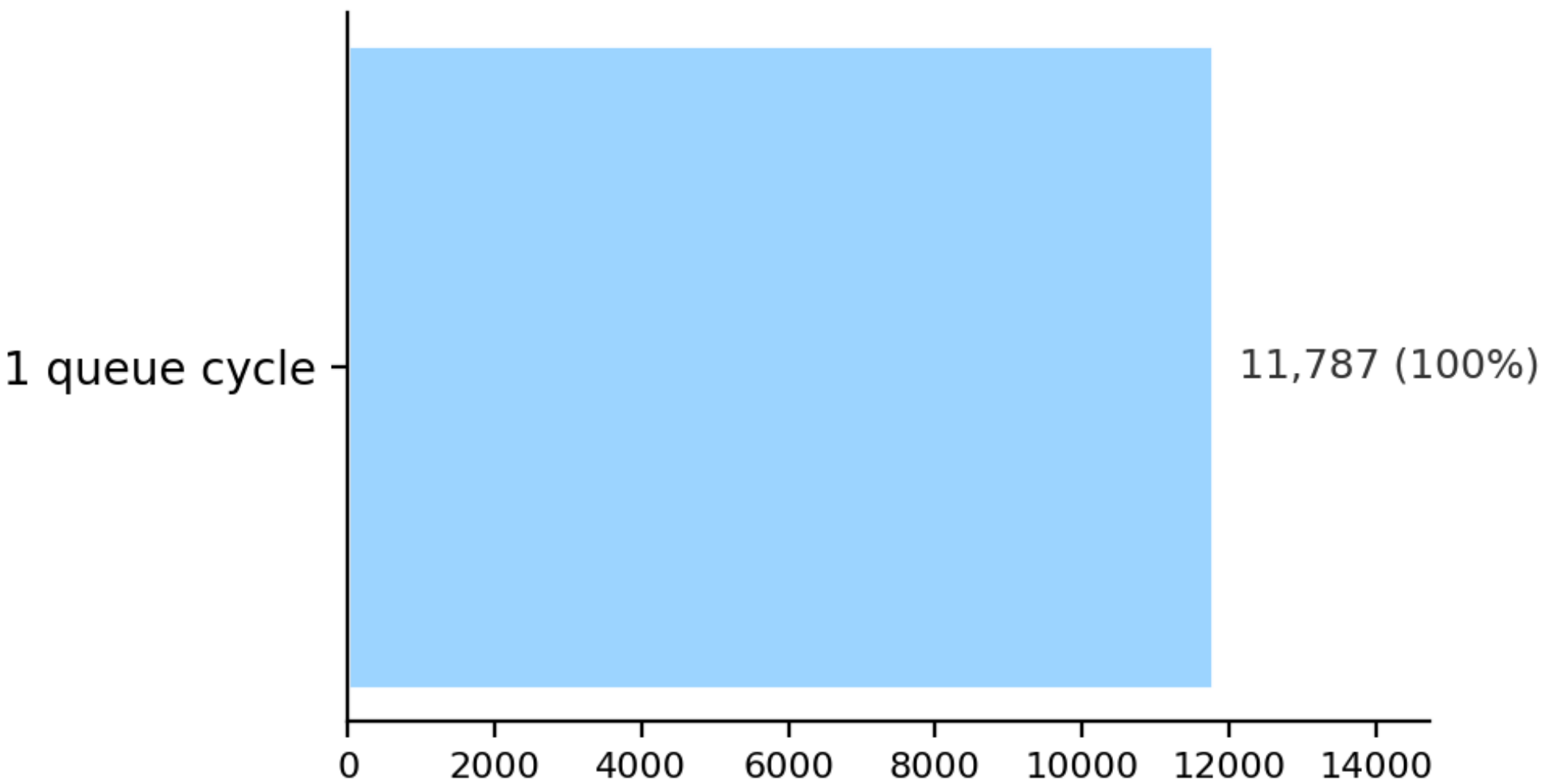
(PR goes back on queue after coming off once)

2 queue cycles — 11,787 PRs

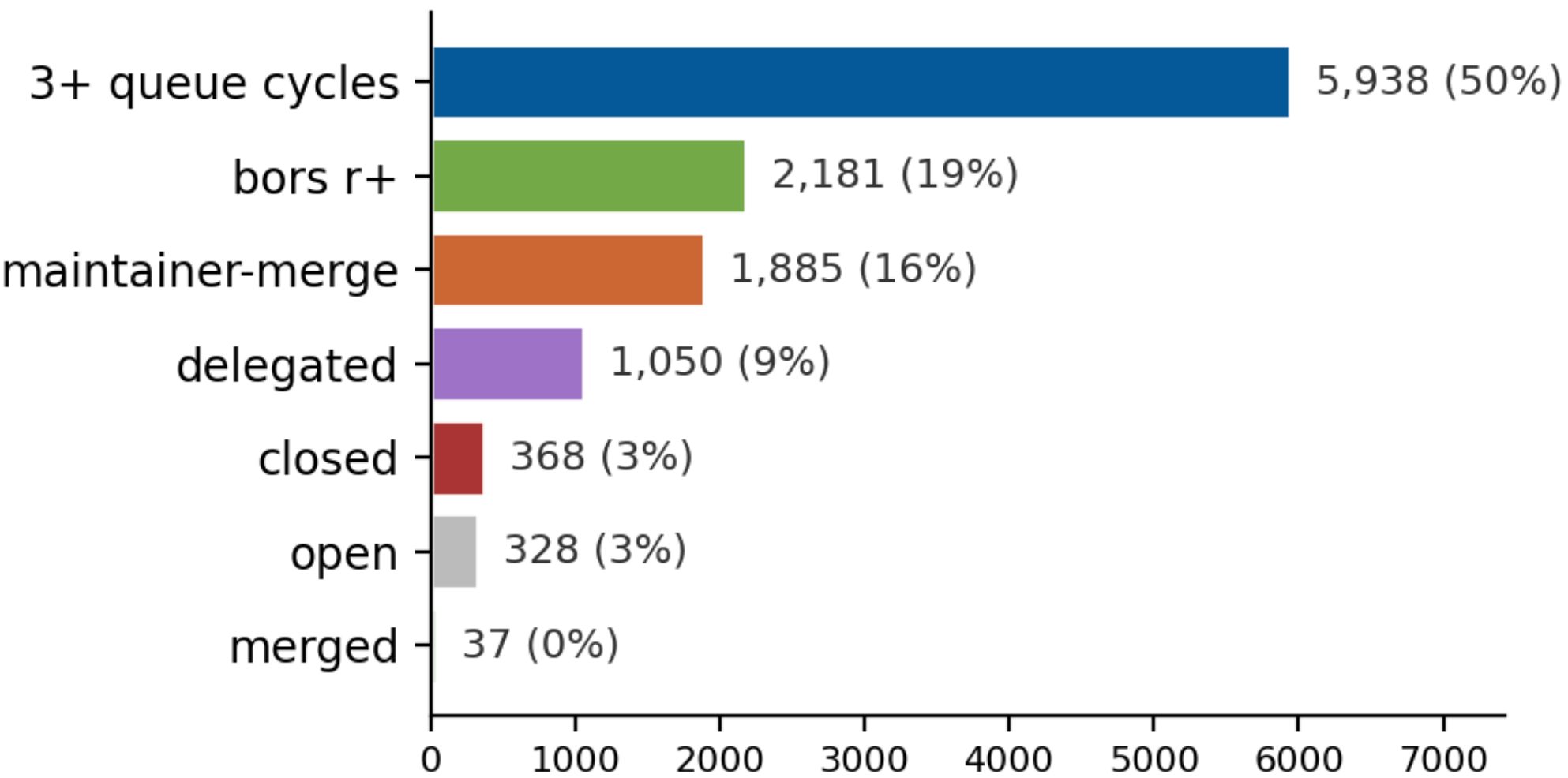


2 queue cycles — flow breakdown

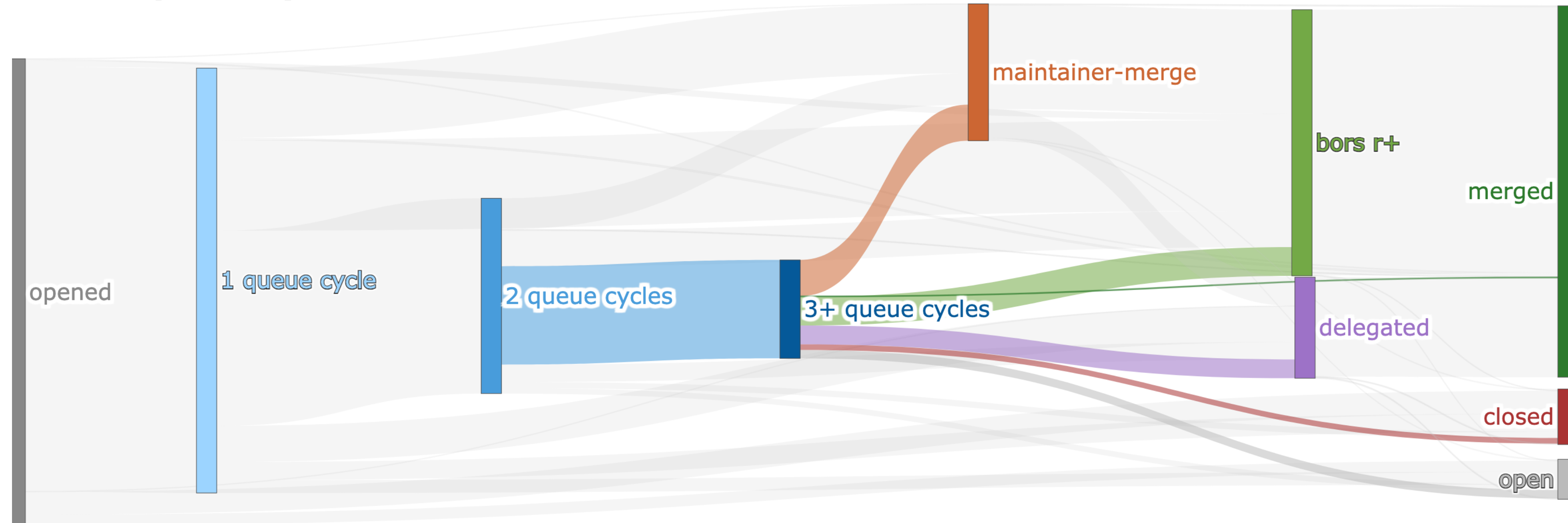
incoming — 11,787 PRs



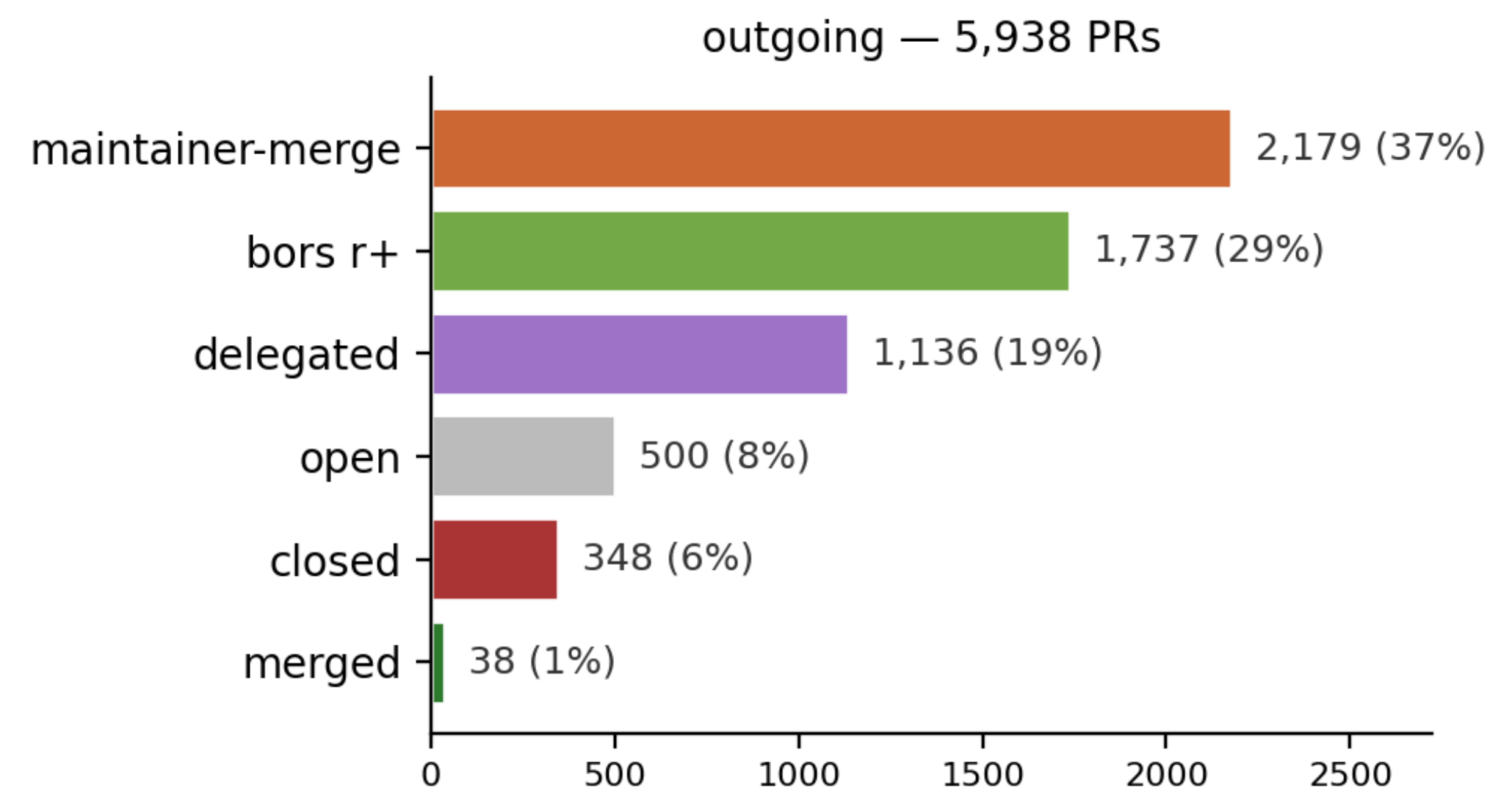
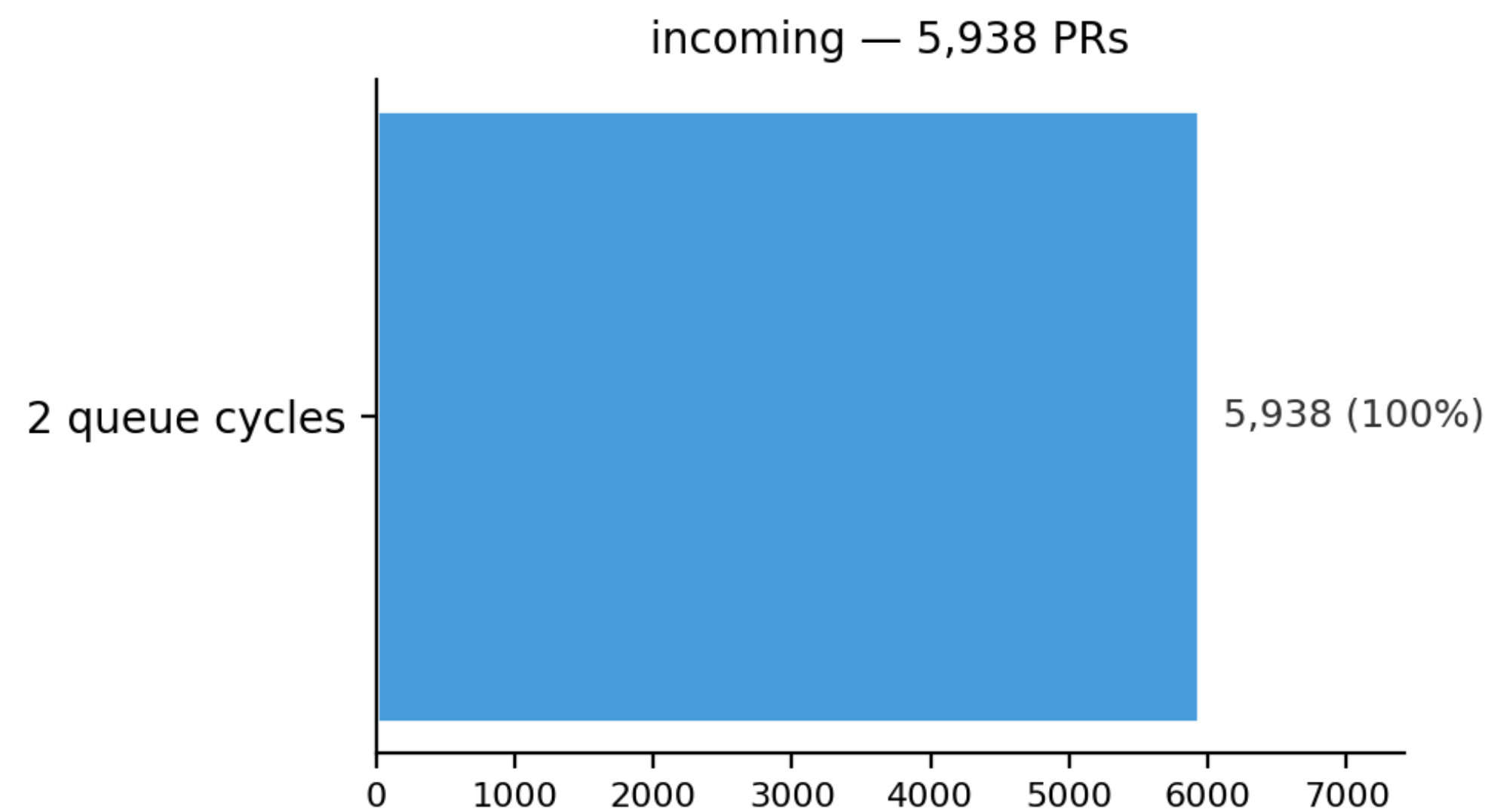
outgoing — 11,787 PRs



3+ queue cycles — 5,938 PRs

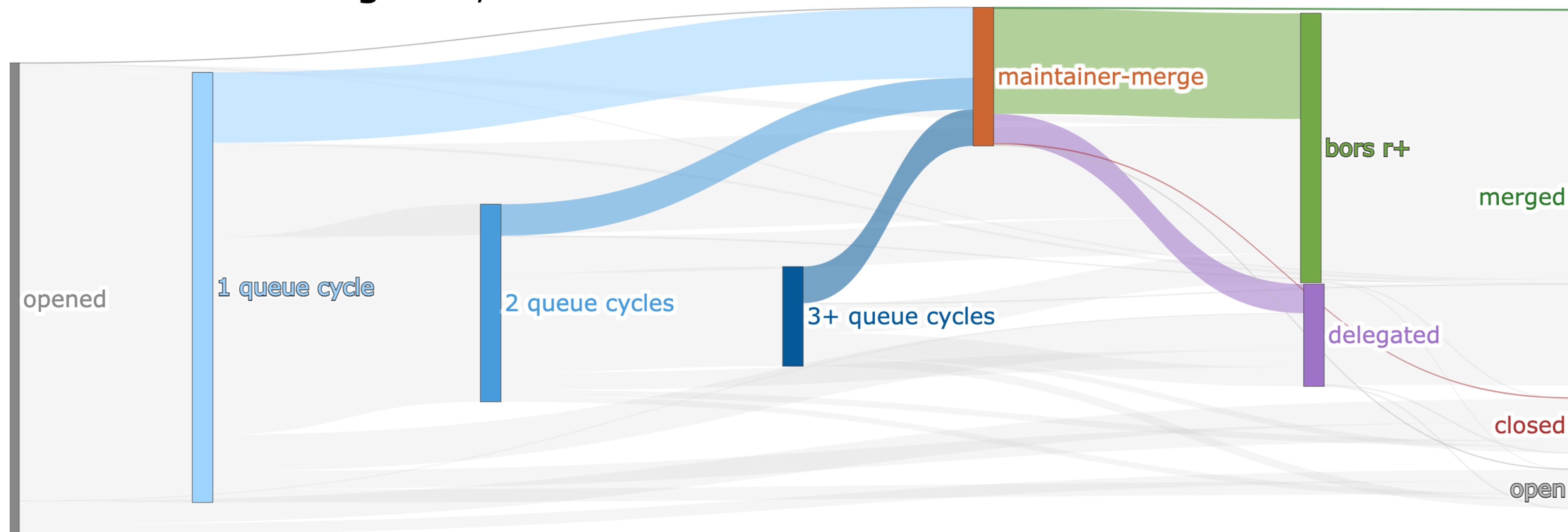


3+ queue cycles — flow breakdown

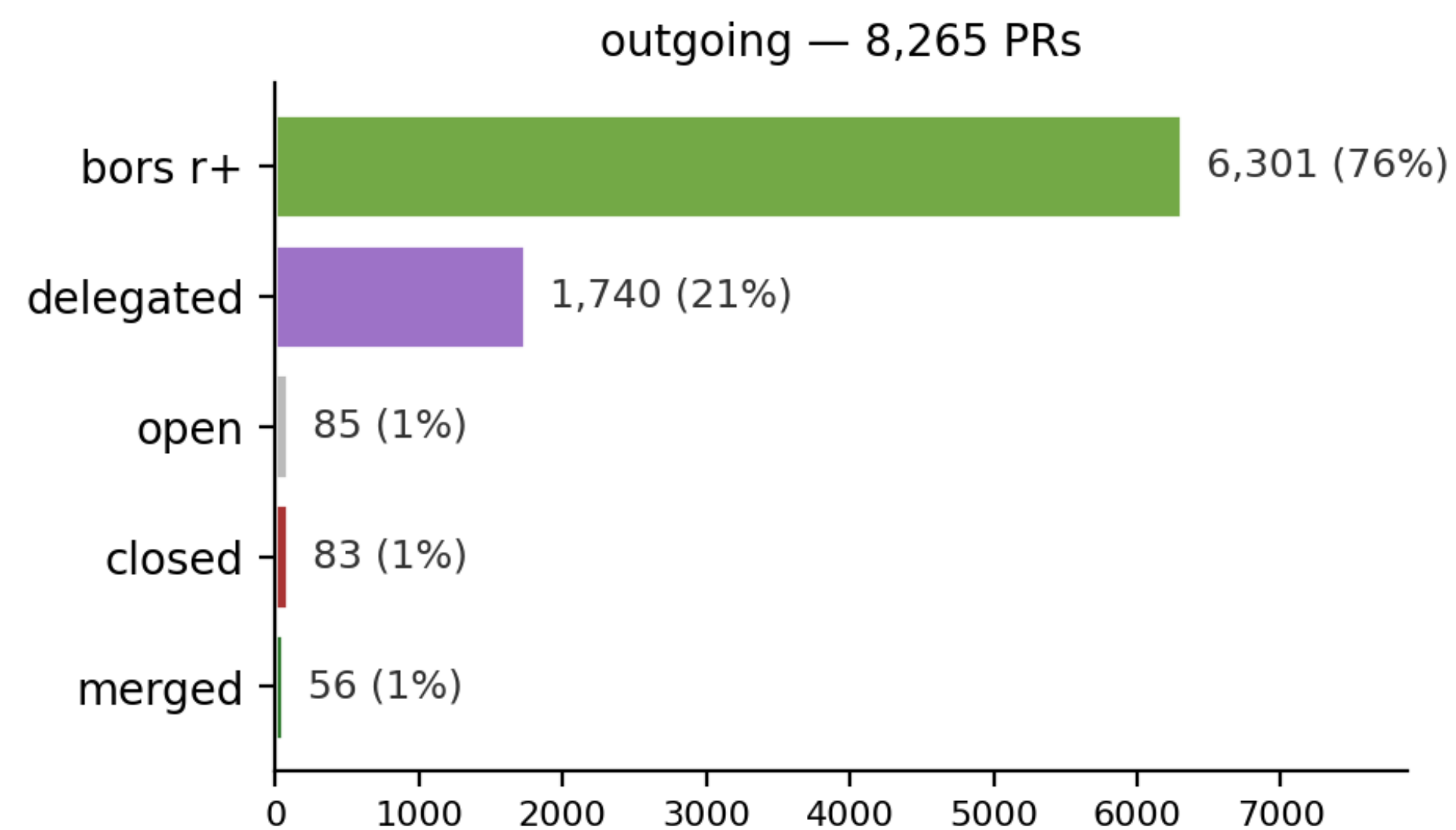
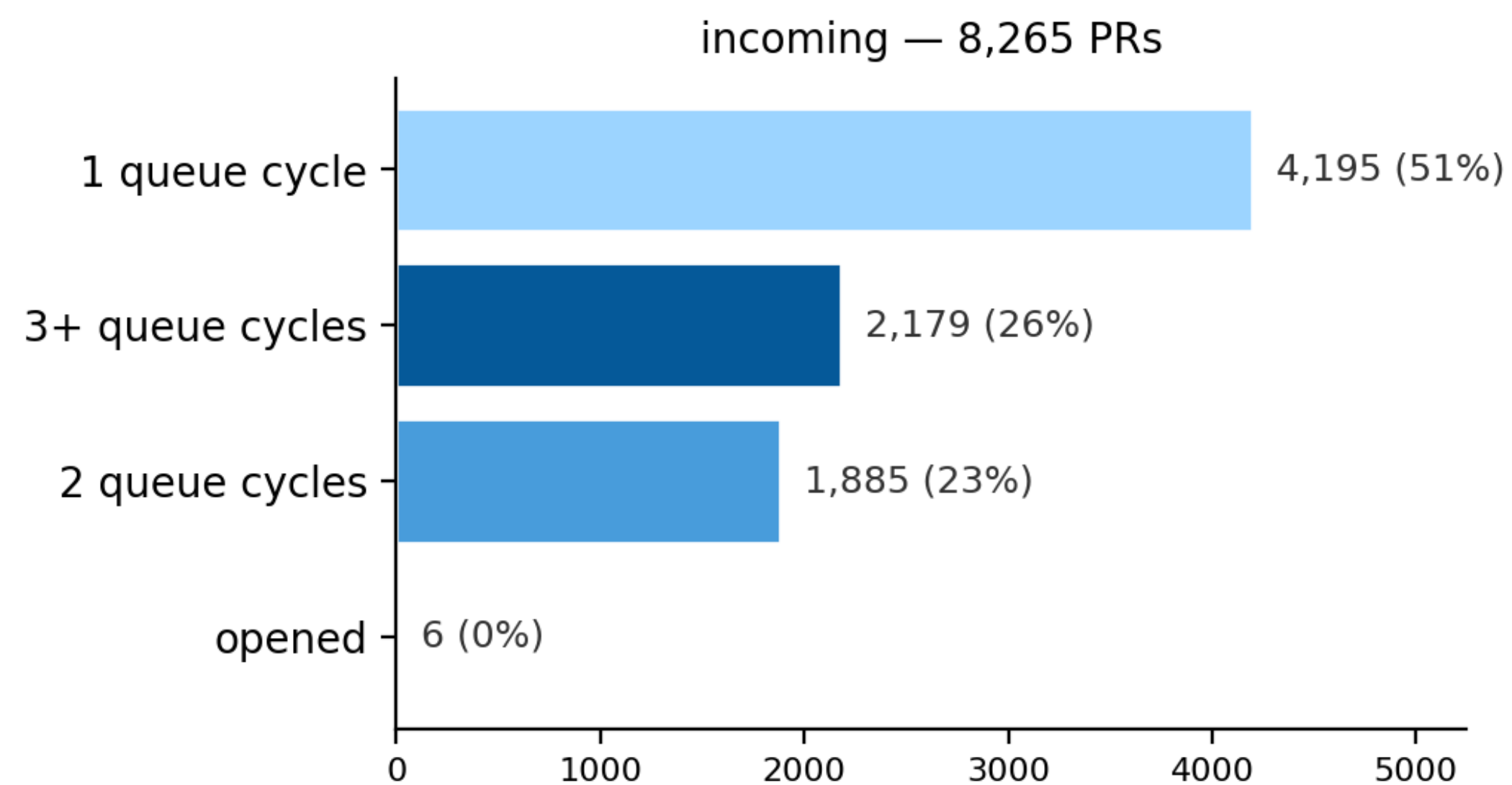


maintainer-merge — 8,265 PRs

(first application of maintainer-merge)

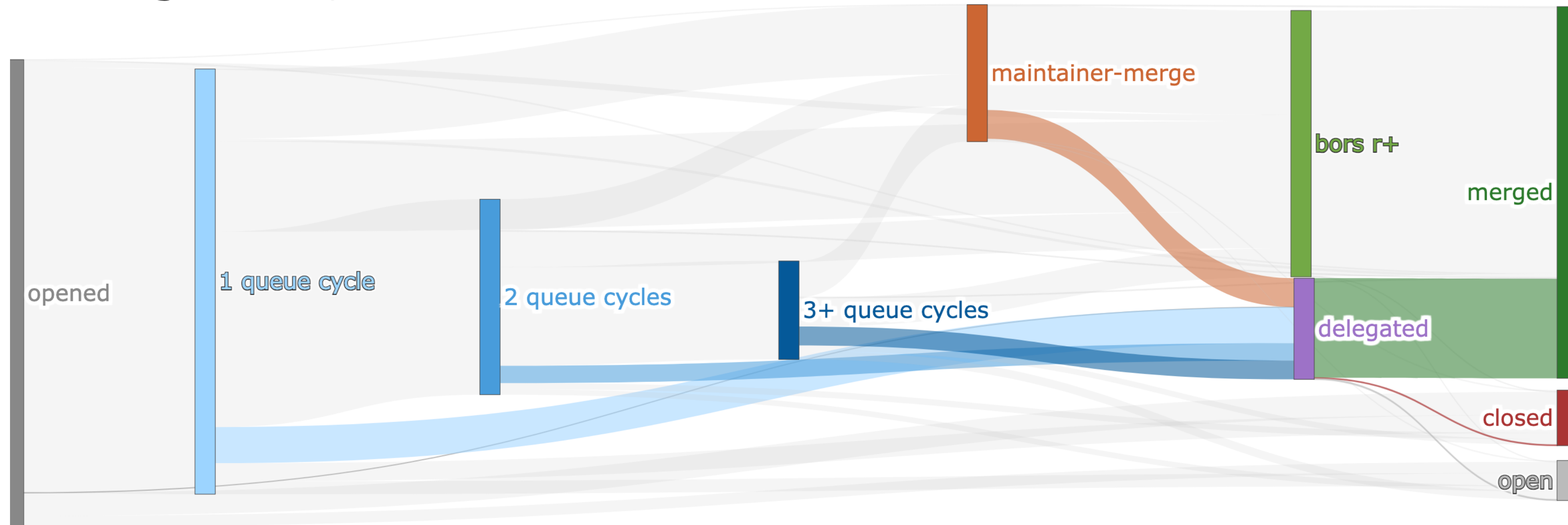


maintainer-merge — flow breakdown



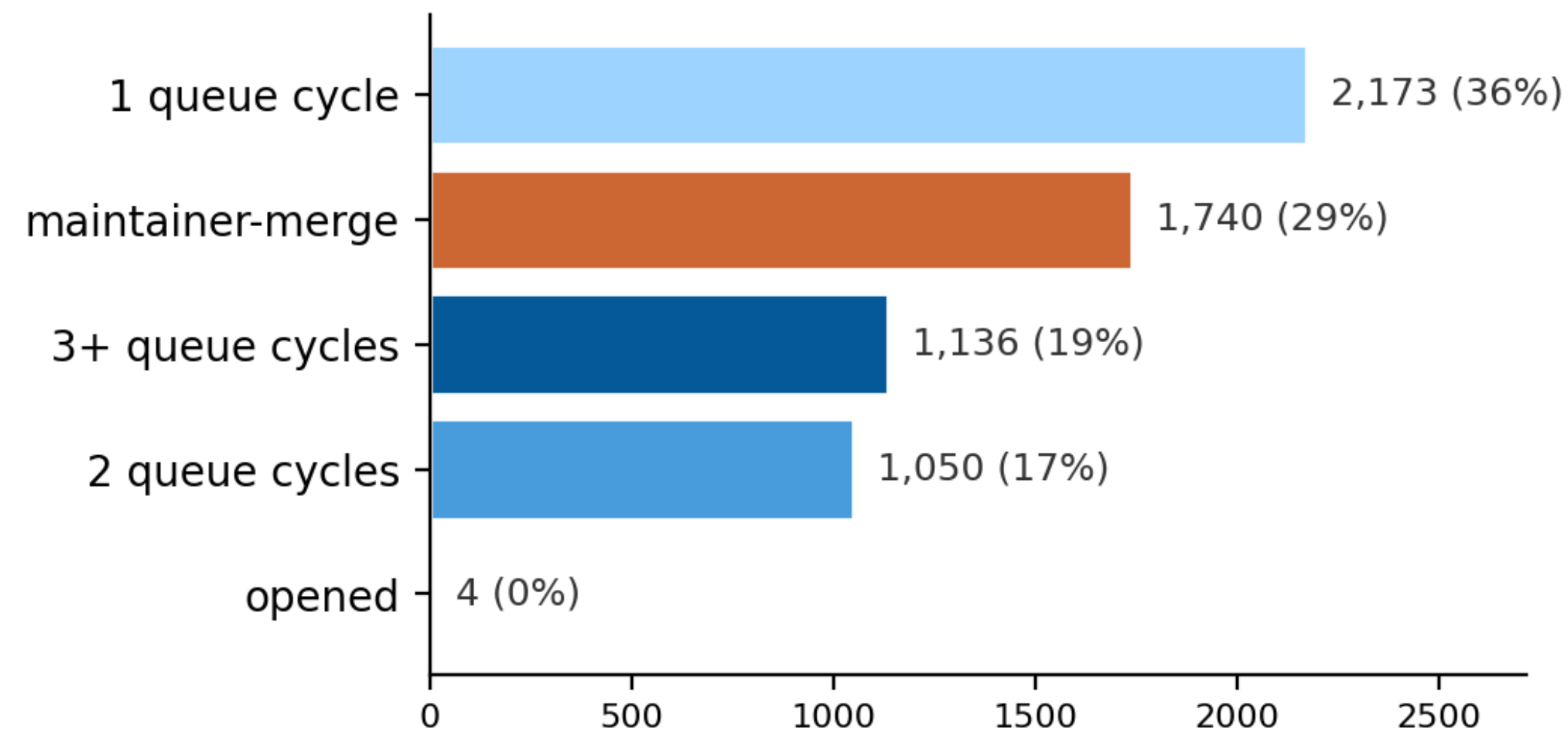
delegated — 6,103 PRs

(first application of delegated)

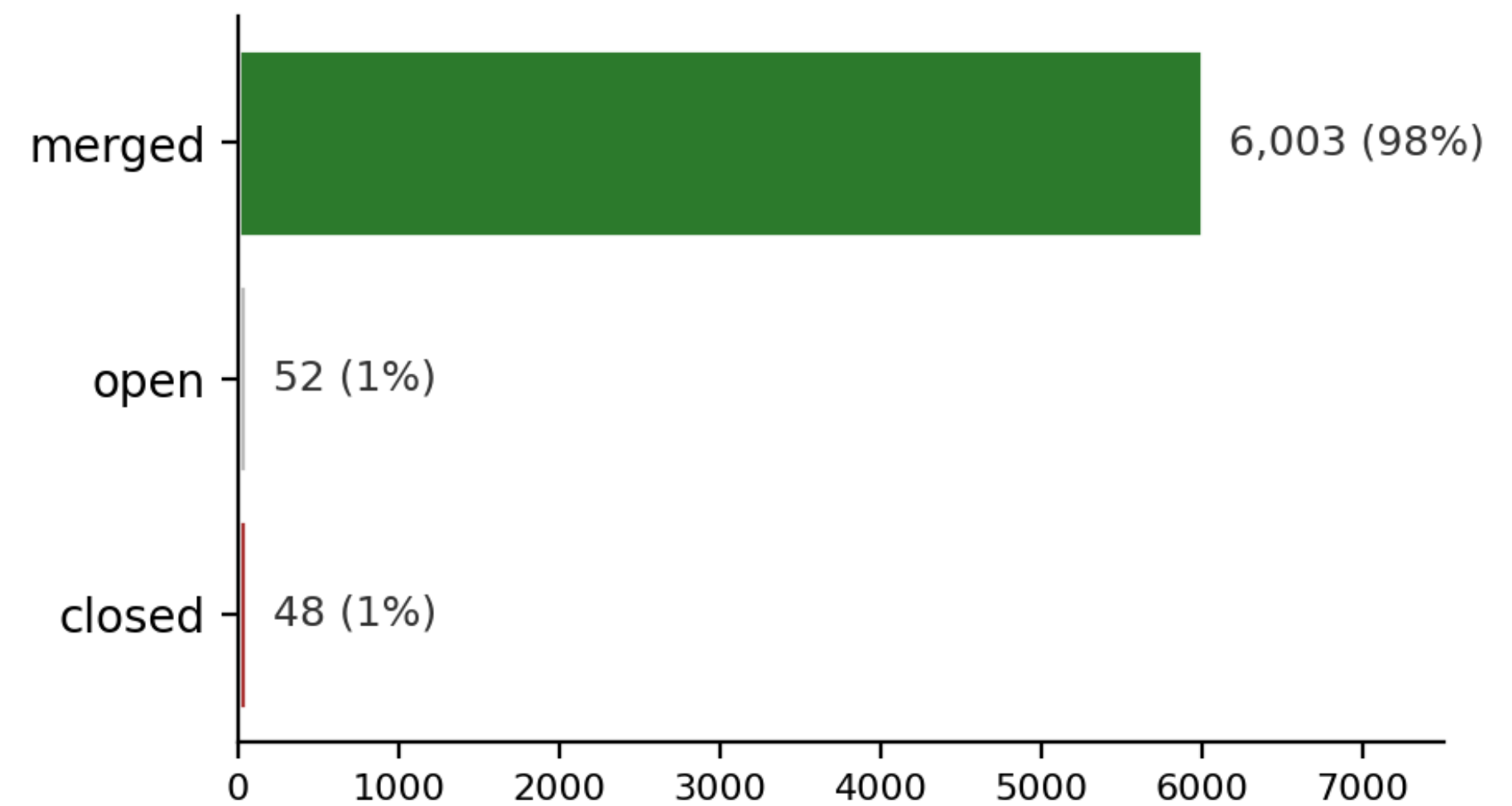


delegated — flow breakdown

incoming — 6,103 PRs

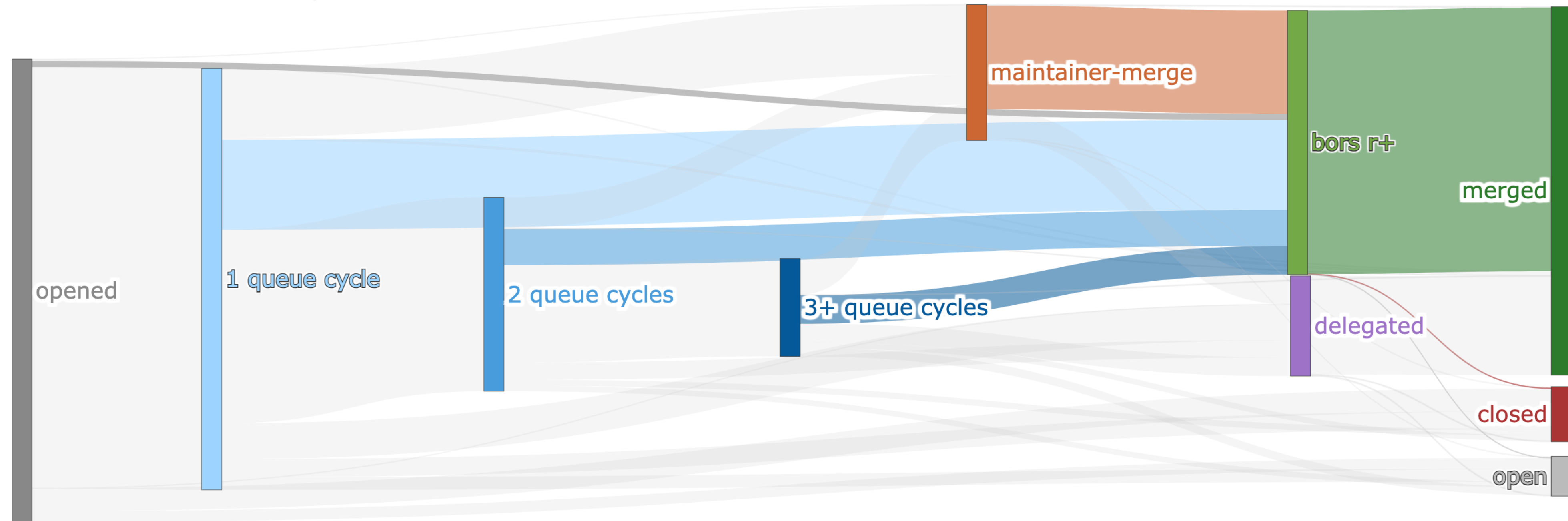


outgoing — 6,103 PRs

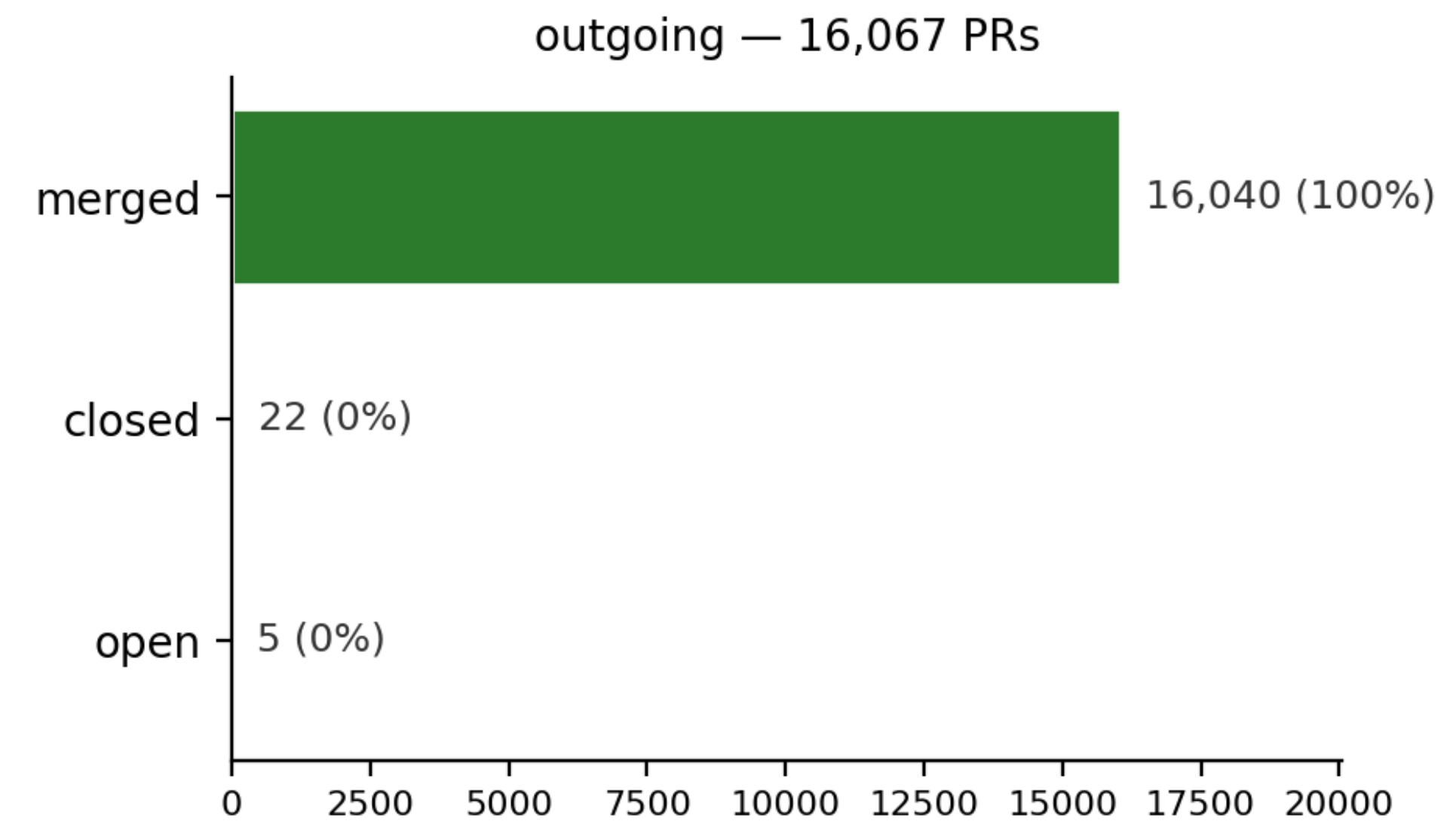
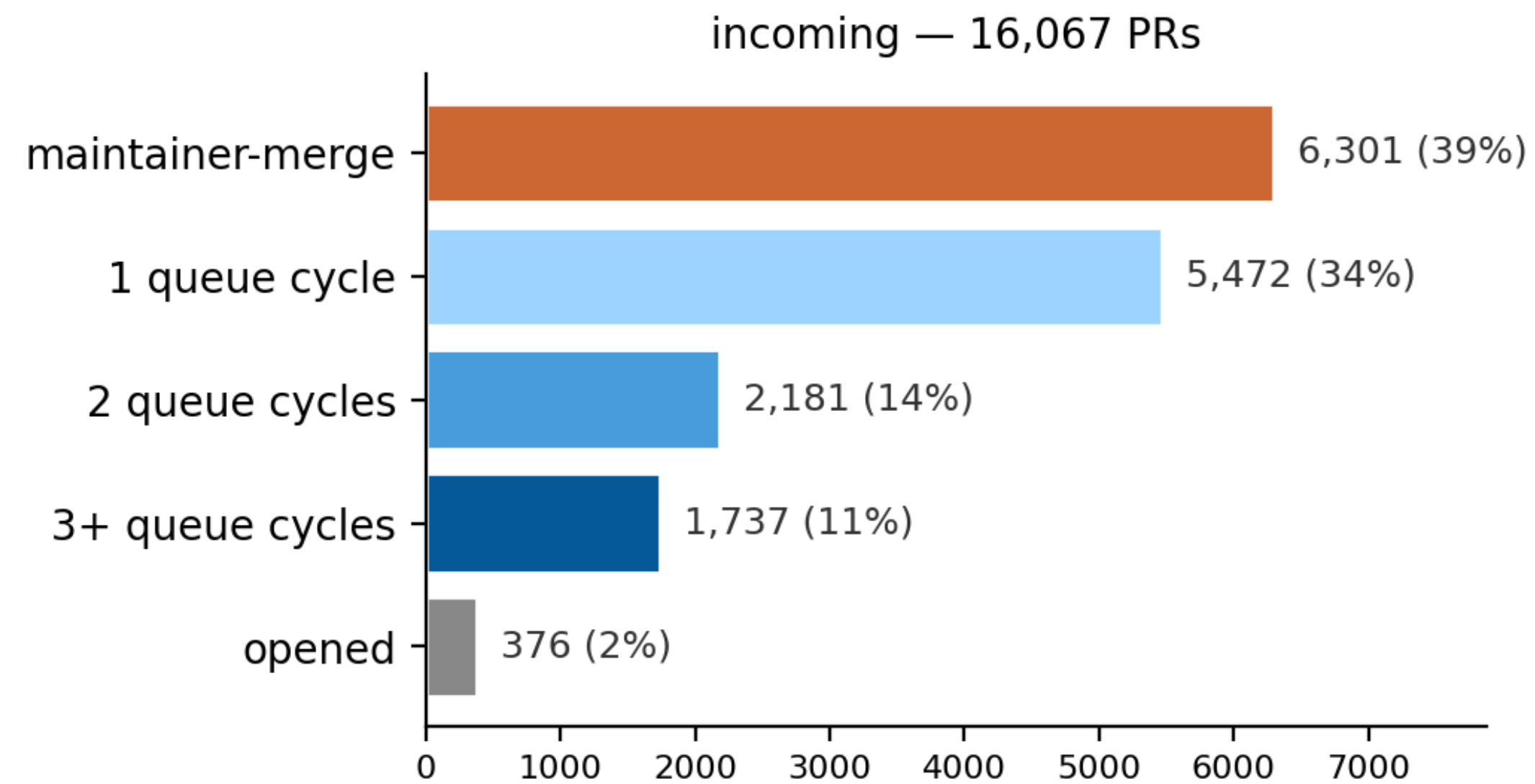


bors r+ — 16,067 PRs

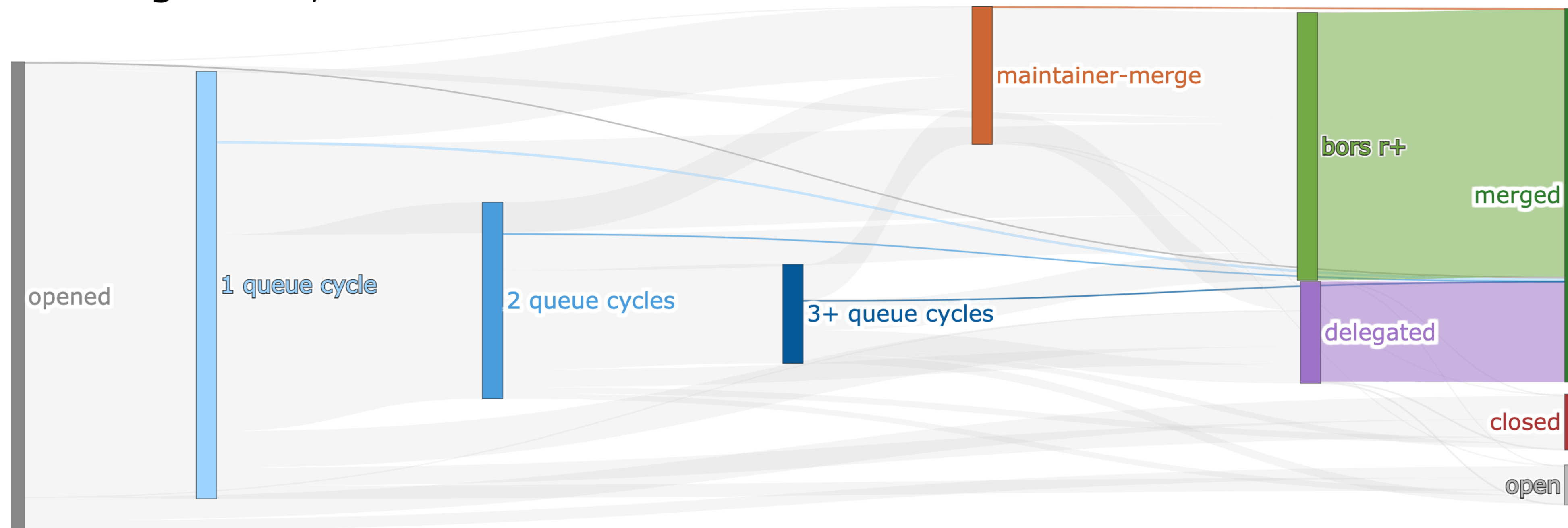
(first application of ready-to-merge (excluding delegated))



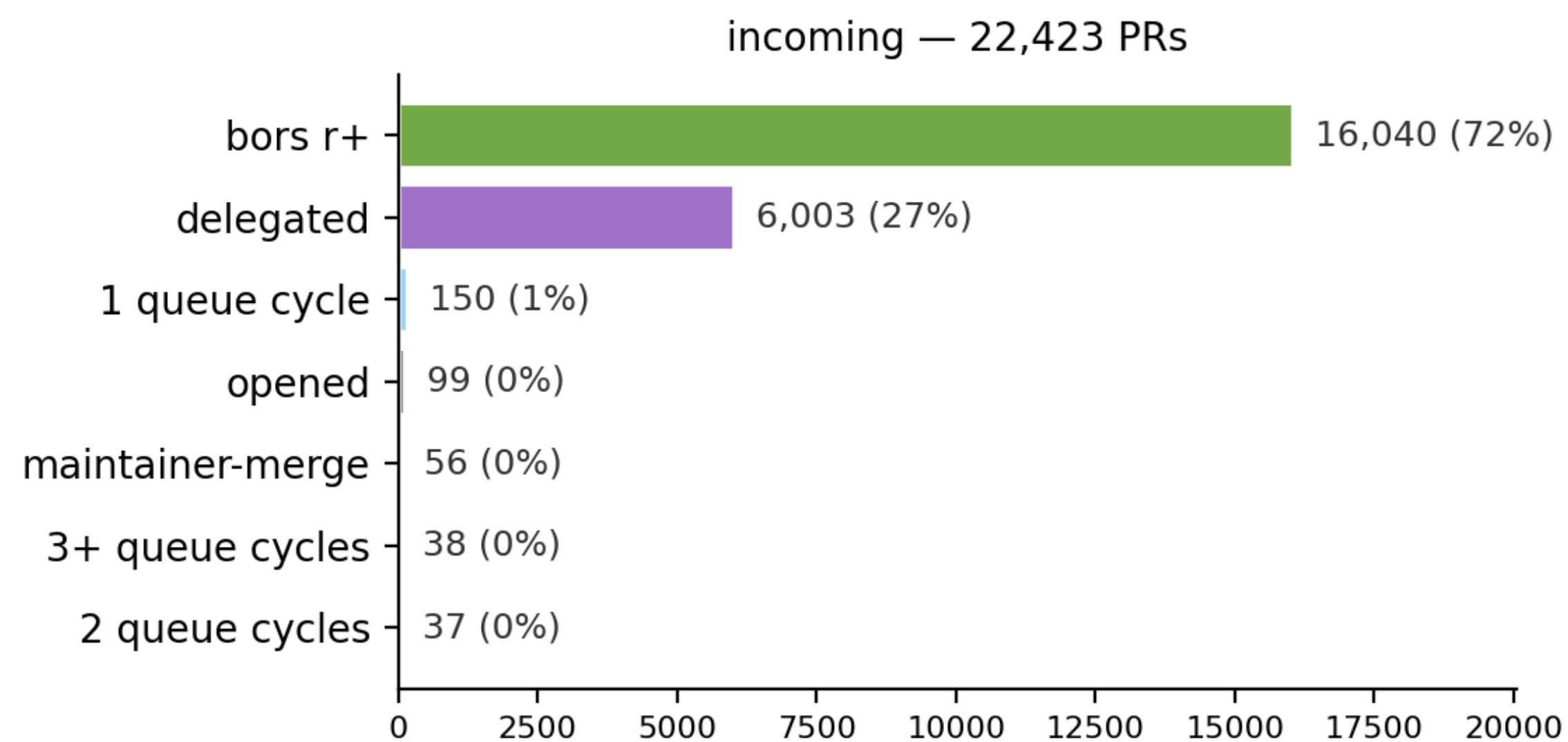
bors r+ — flow breakdown



merged — 22,423 PRs



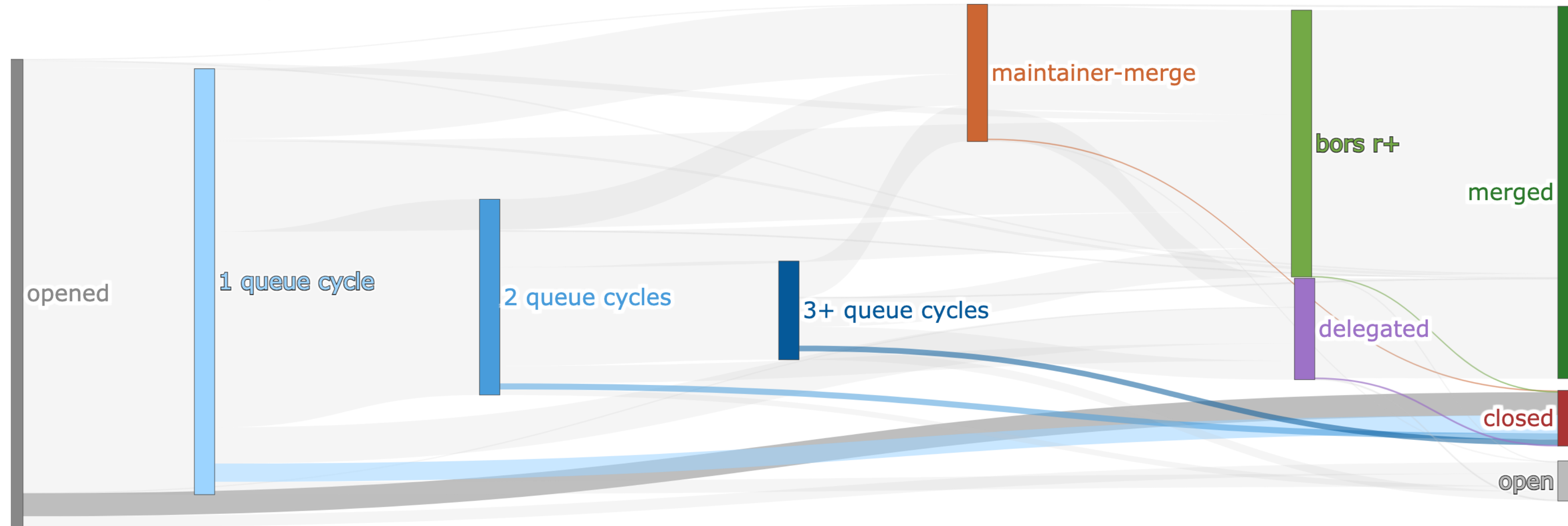
merged — flow breakdown



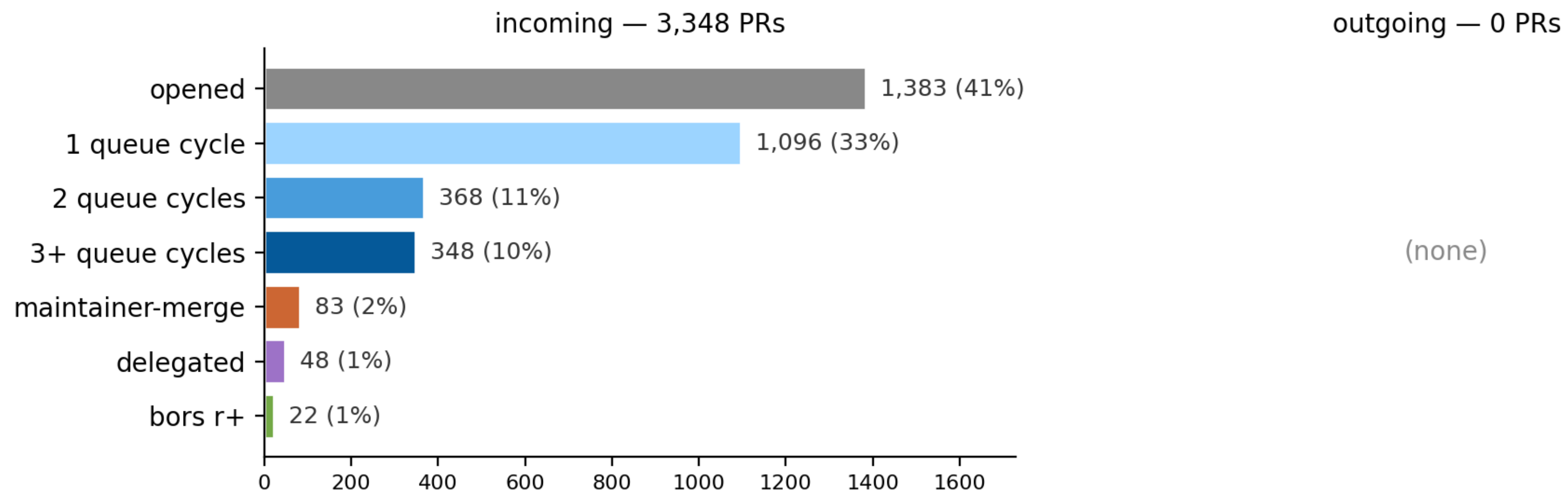
outgoing — 0 PRs

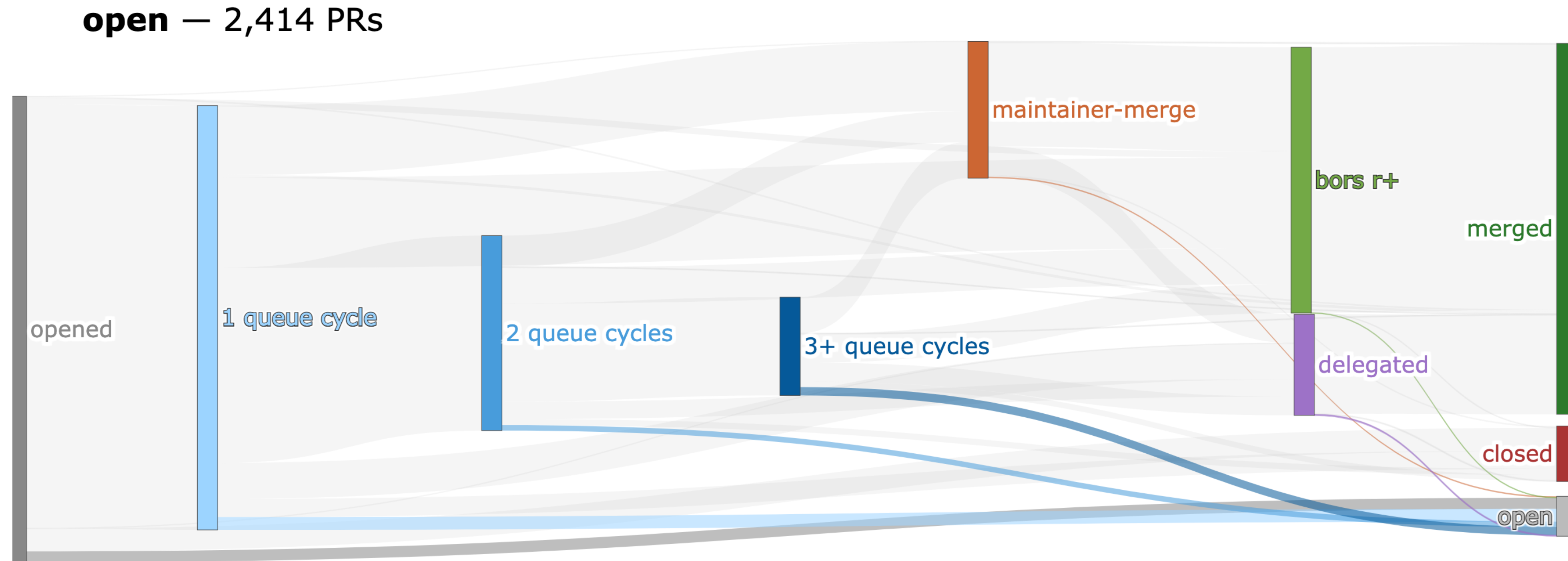
(none)

closed — 3,348 PRs

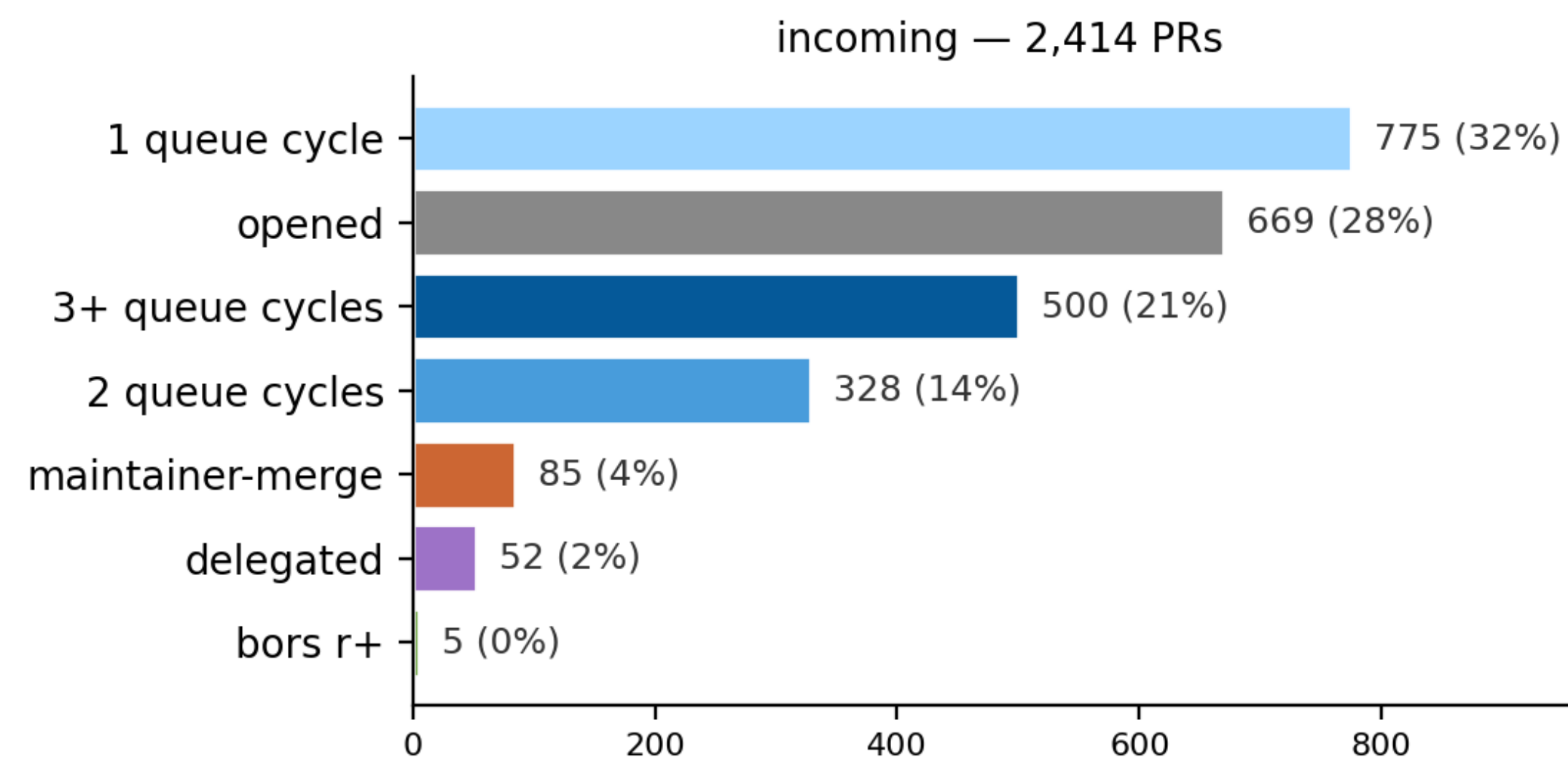


closed — flow breakdown





open — flow breakdown



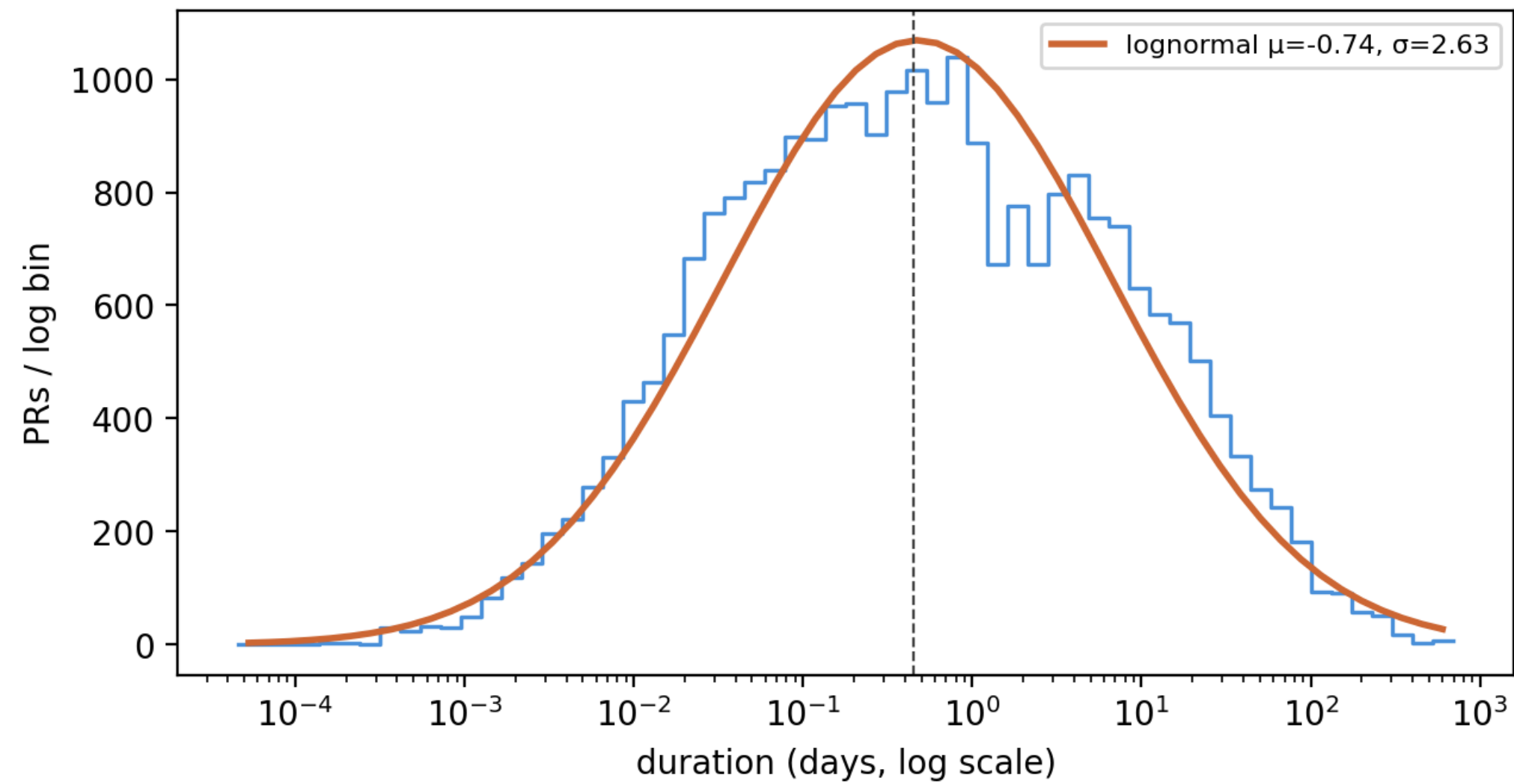
outgoing — 0 PRs

(none)

How long does each stage take?

All (10.8 h)

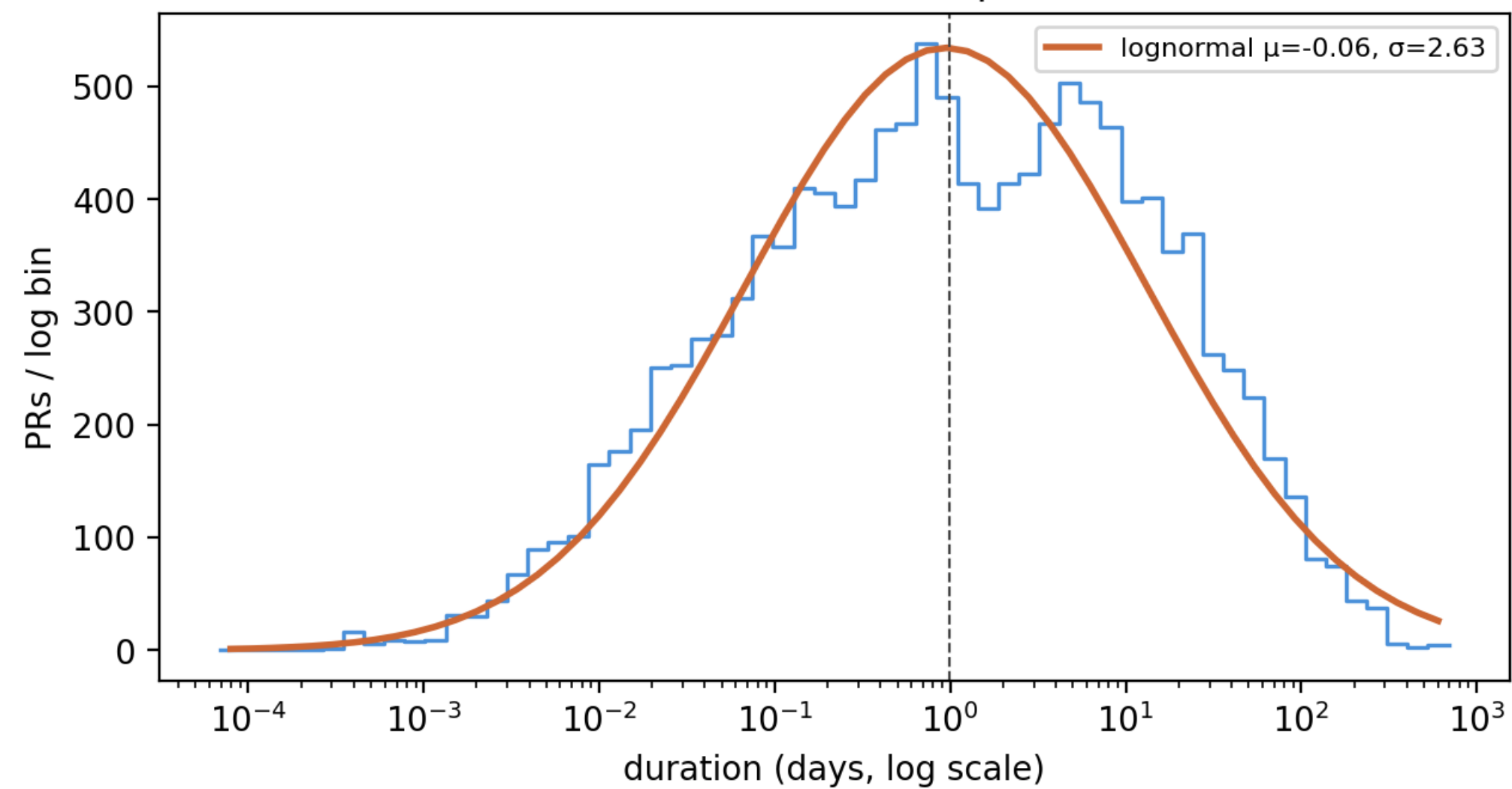
open → first review
n=25648, median=0.45d, p90=16.62d



Open -> First review

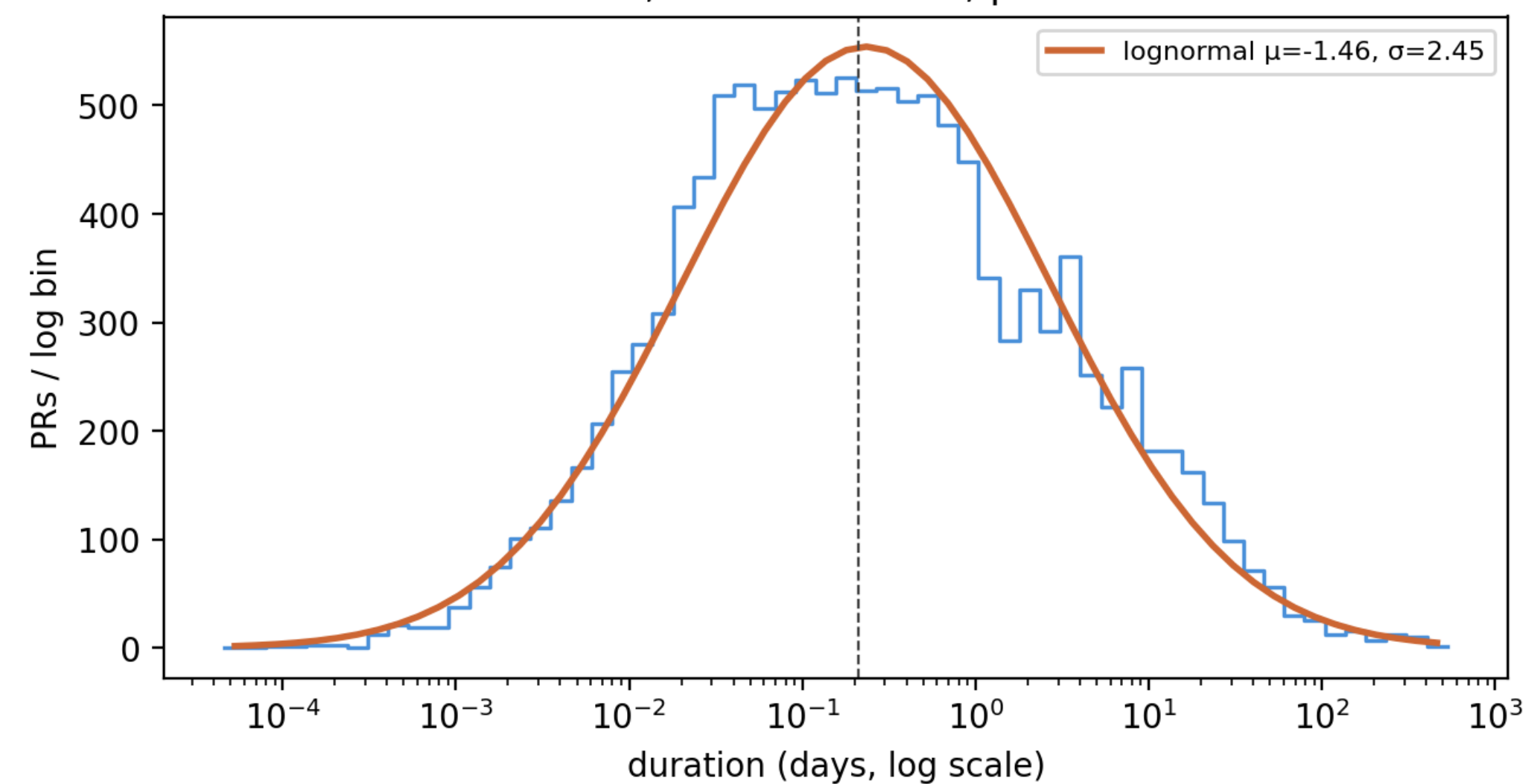
feat (23.8 h)

open → first review
n=13092, median=0.99d, p90=26.91d



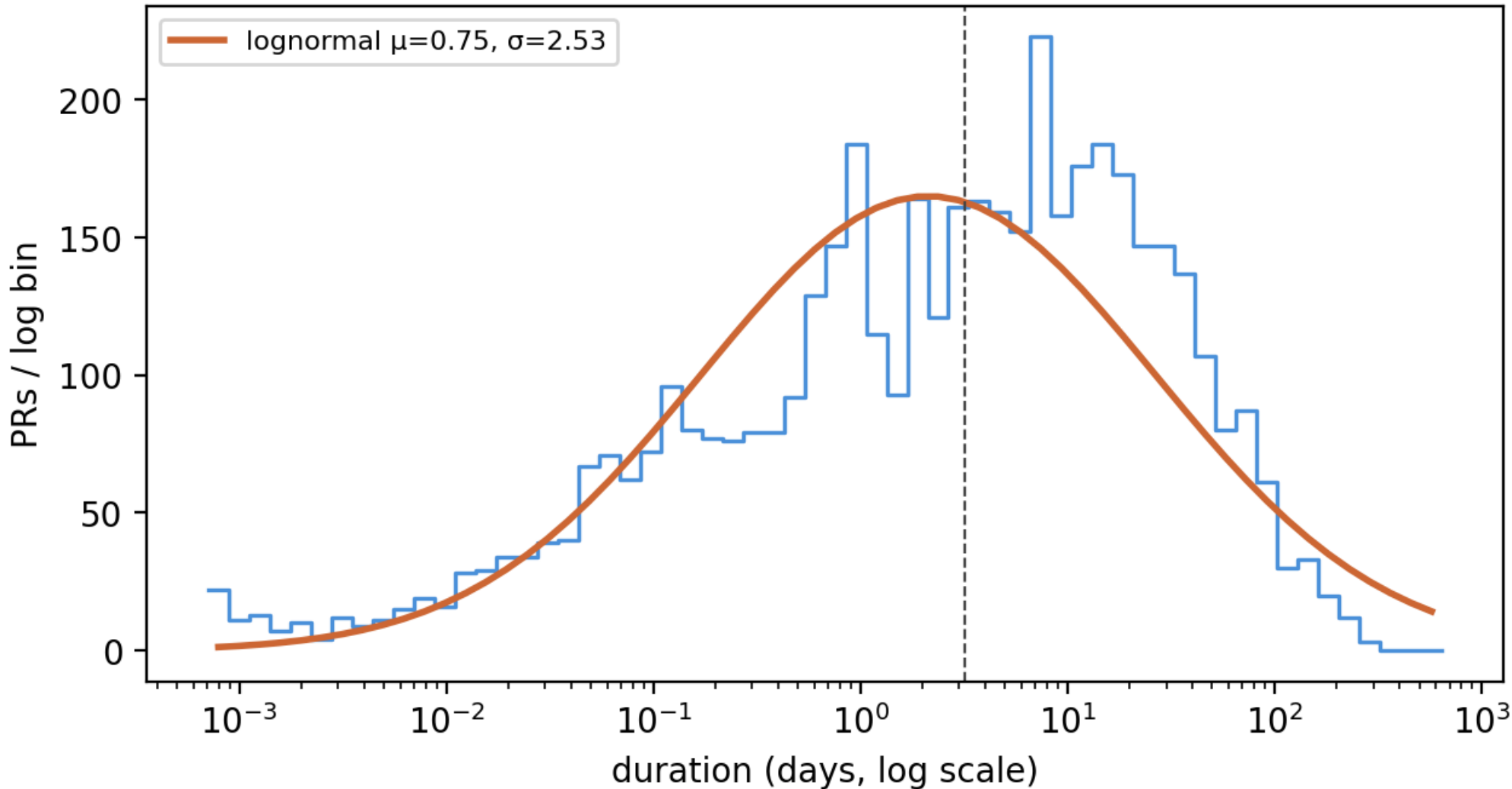
non-feat (5.0 h)

open → first review
n=12556, median=0.21d, p90=6.93d



All (3.2d=76h)

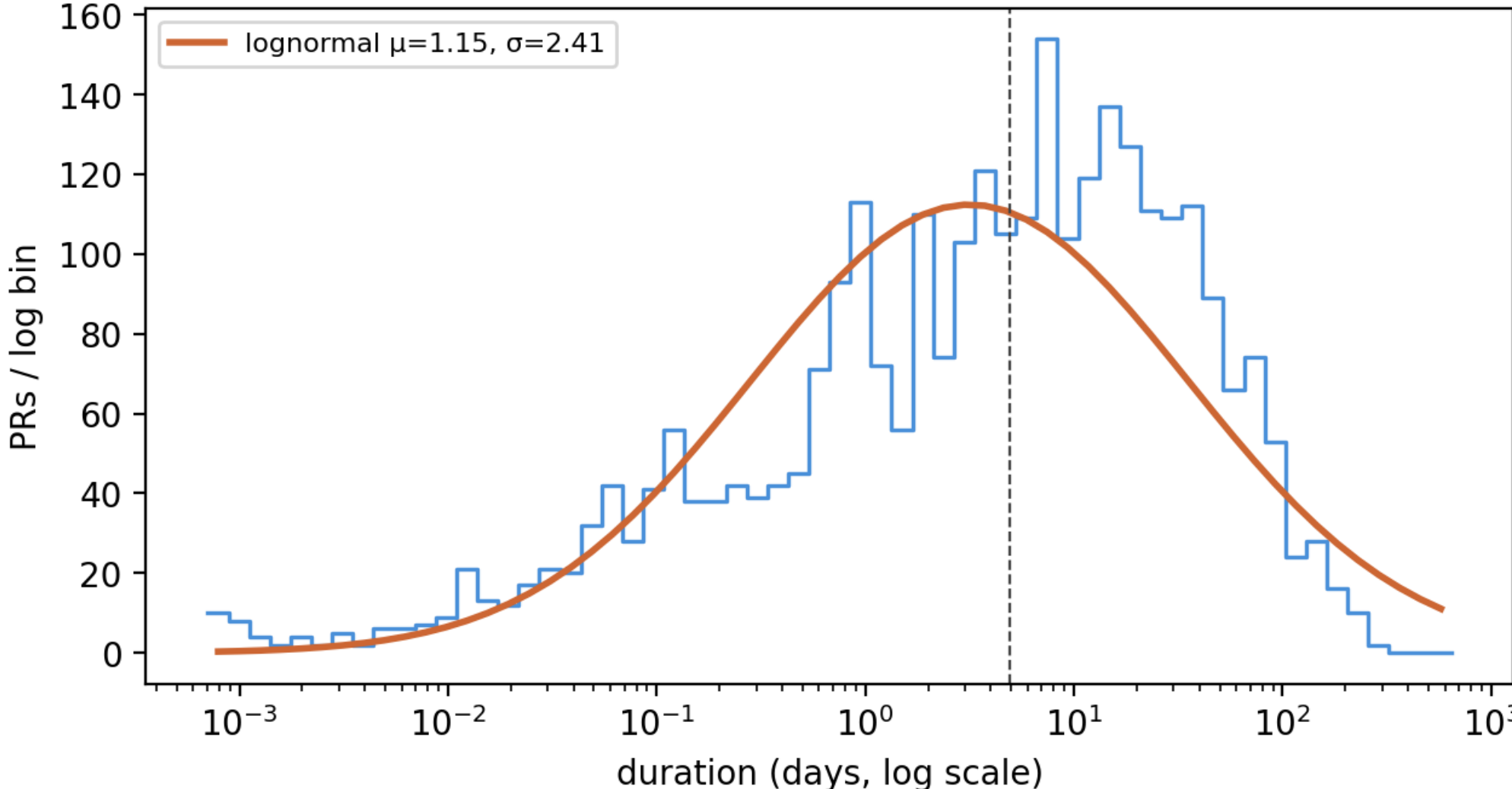
first review → maintainer-merge (iterated, ≥ 60 s)
n=4571, median=3.18d, p90=39.67d



First review -> maintainer merge

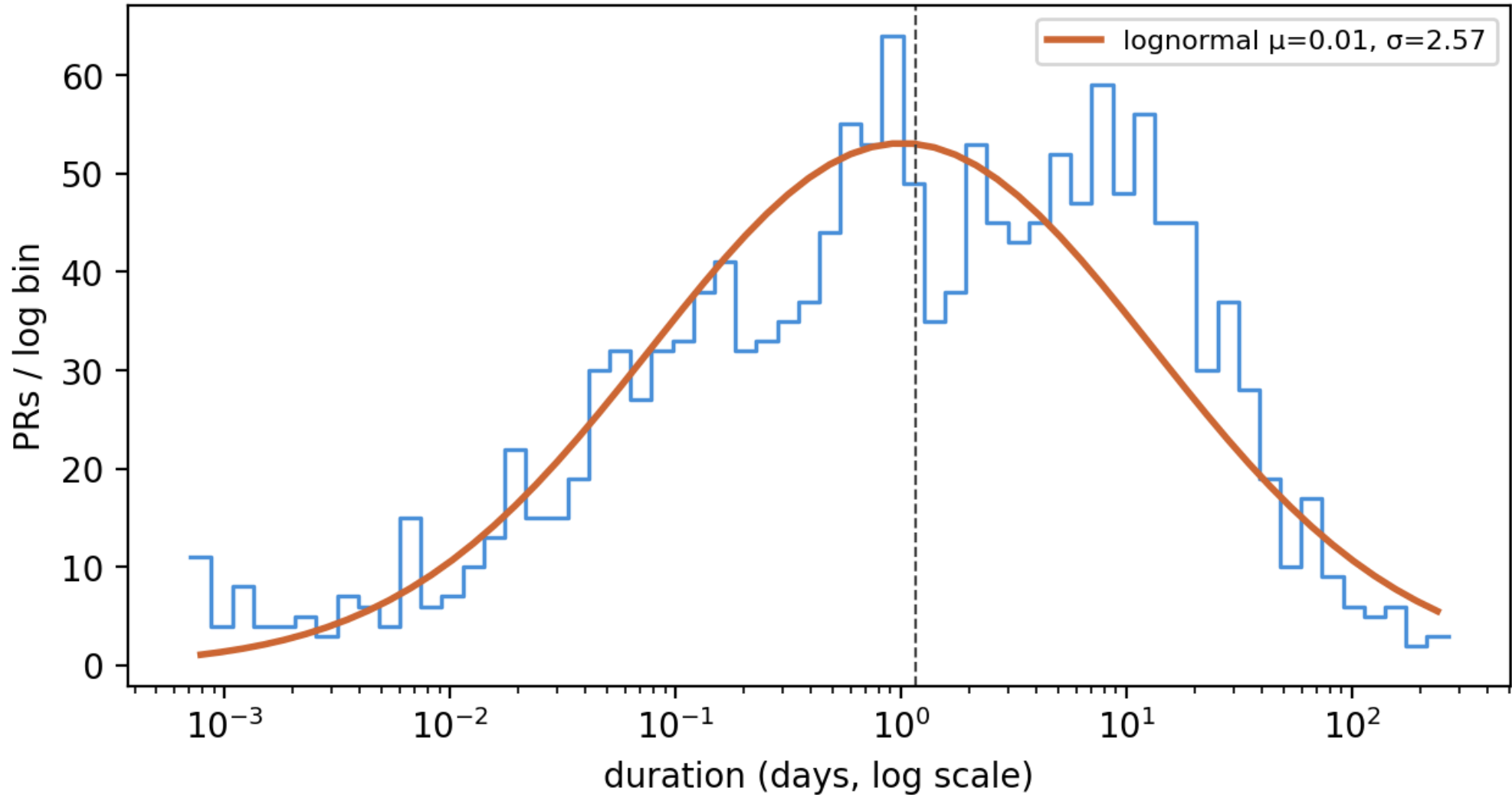
feat (4.9d)

first review → maintainer-merge (iterated, ≥ 60 s)
n=2975, median=4.93d, p90=49.05d



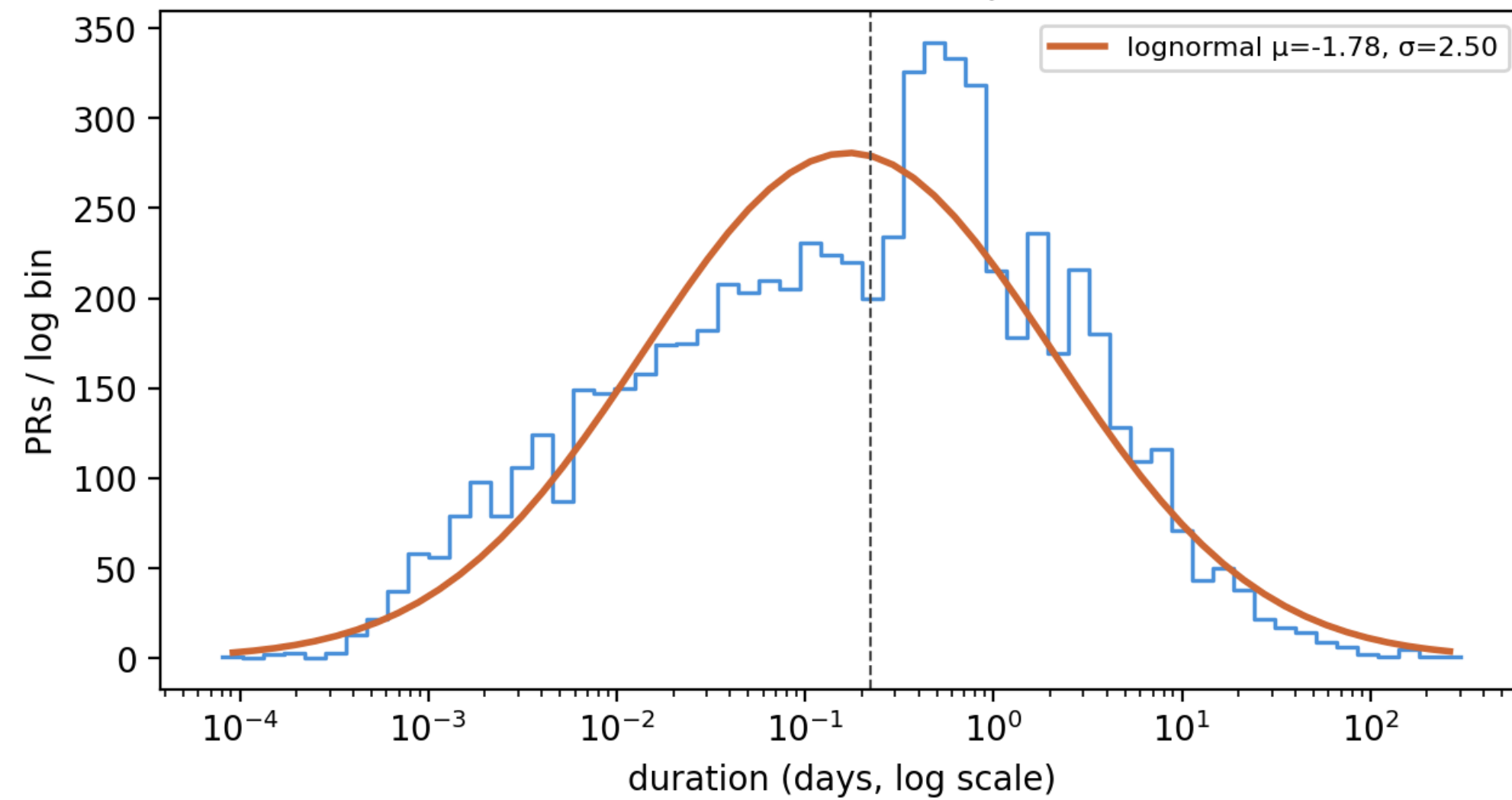
non-feat (1.2d=28h)

first review → maintainer-merge (iterated, ≥ 60 s)
n=1596, median=1.16d, p90=22.03d



All (5.3h)

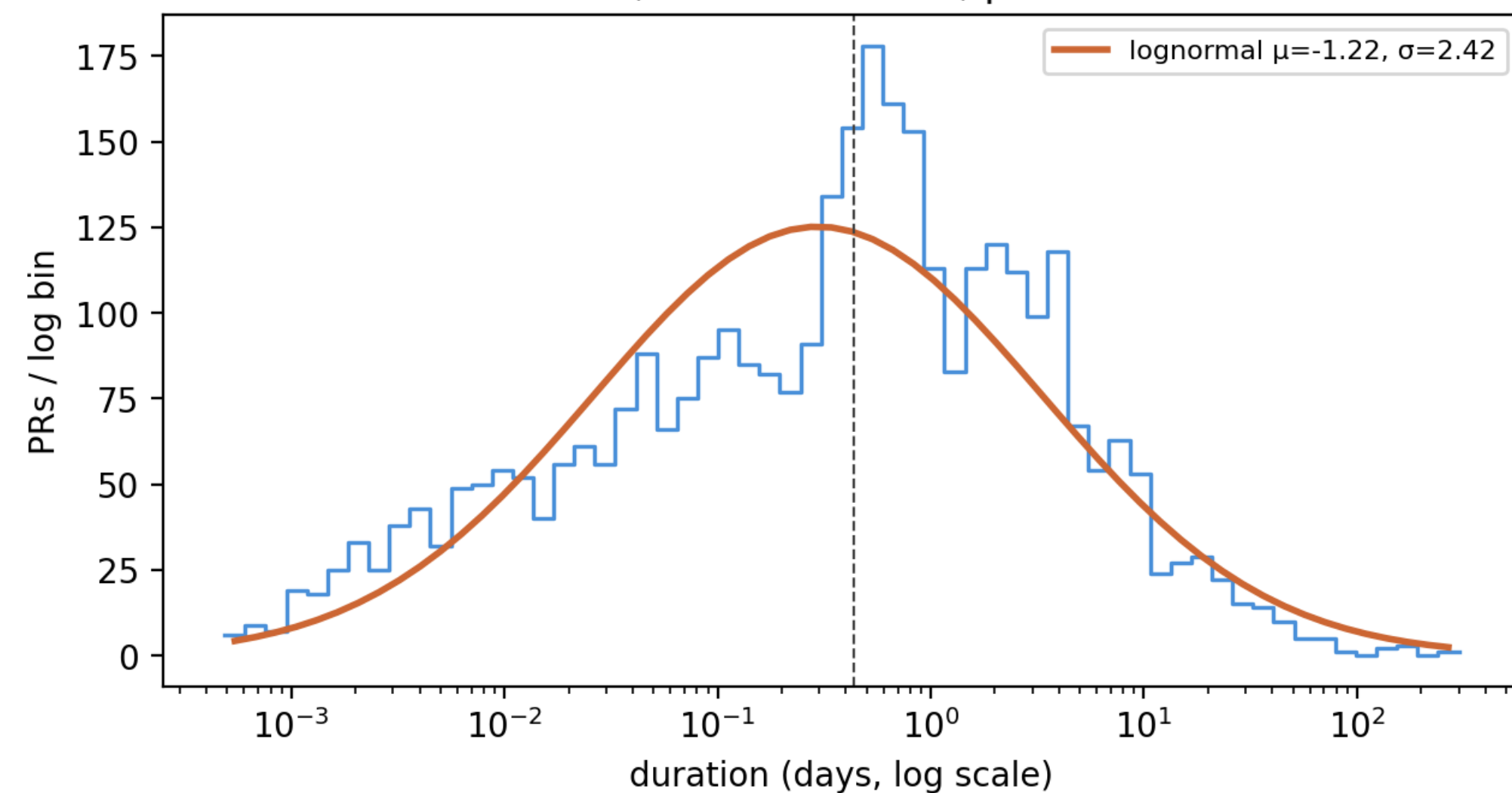
MM → ready-to-merge
n=6985, median=0.22d, p90=3.80d



maintainer merge -> ready-to-merge
(both delegated and undelegated)

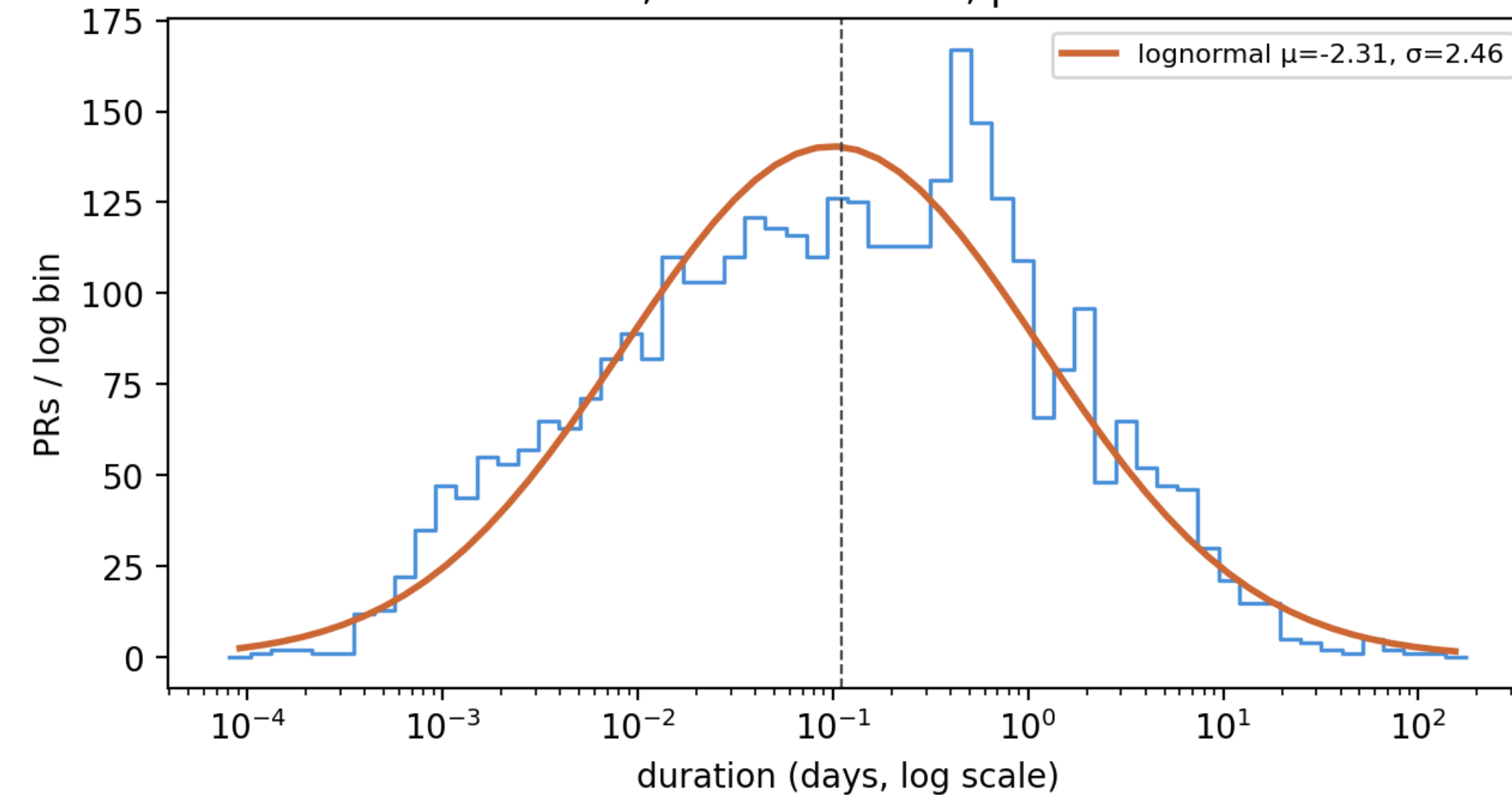
feat (10.3h)

MM → ready-to-merge
n=3424, median=0.43d, p90=5.22d



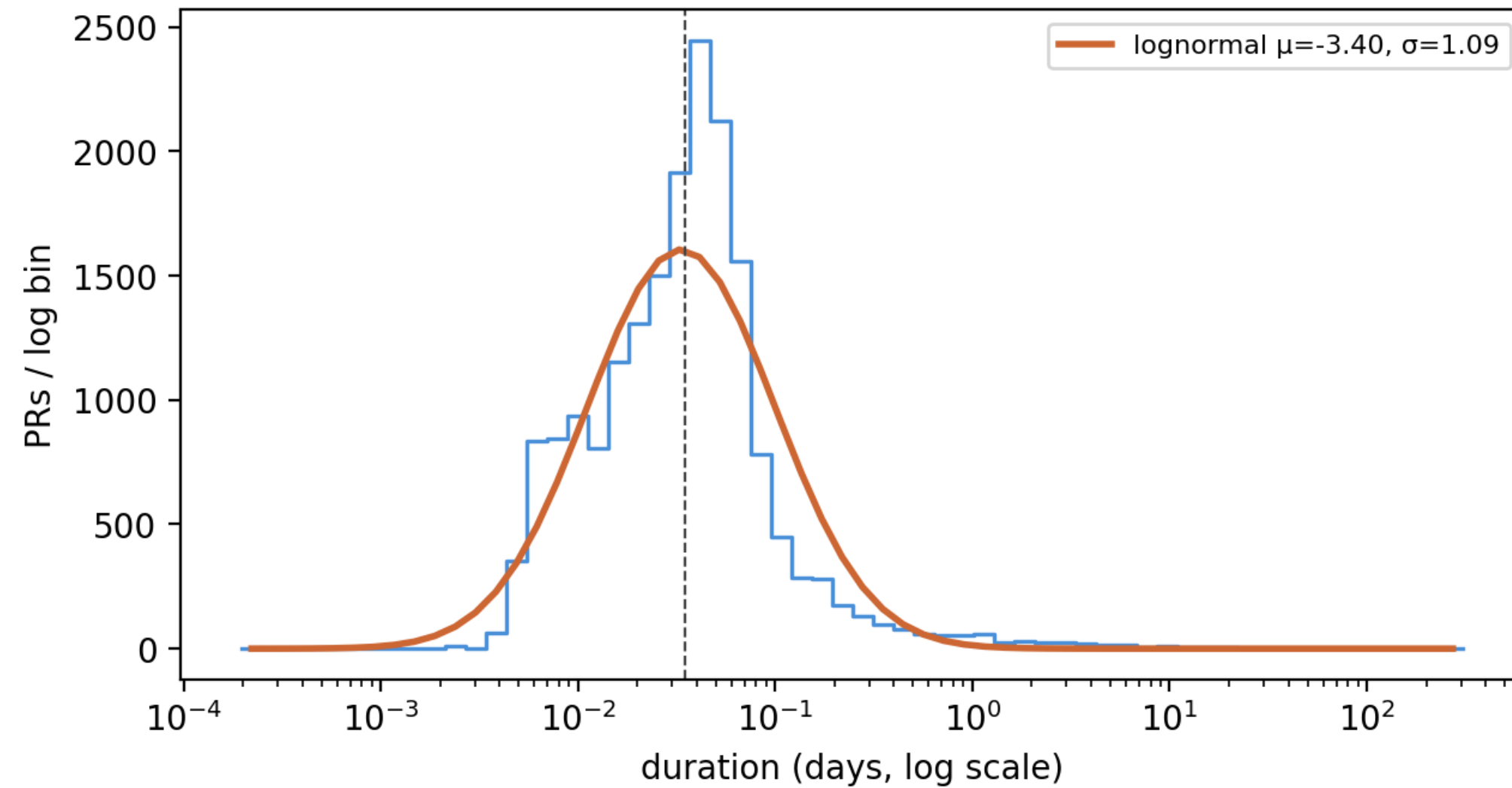
non-feat (2.6h)

MM → ready-to-merge
n=3561, median=0.11d, p90=2.25d



All (50 mins)

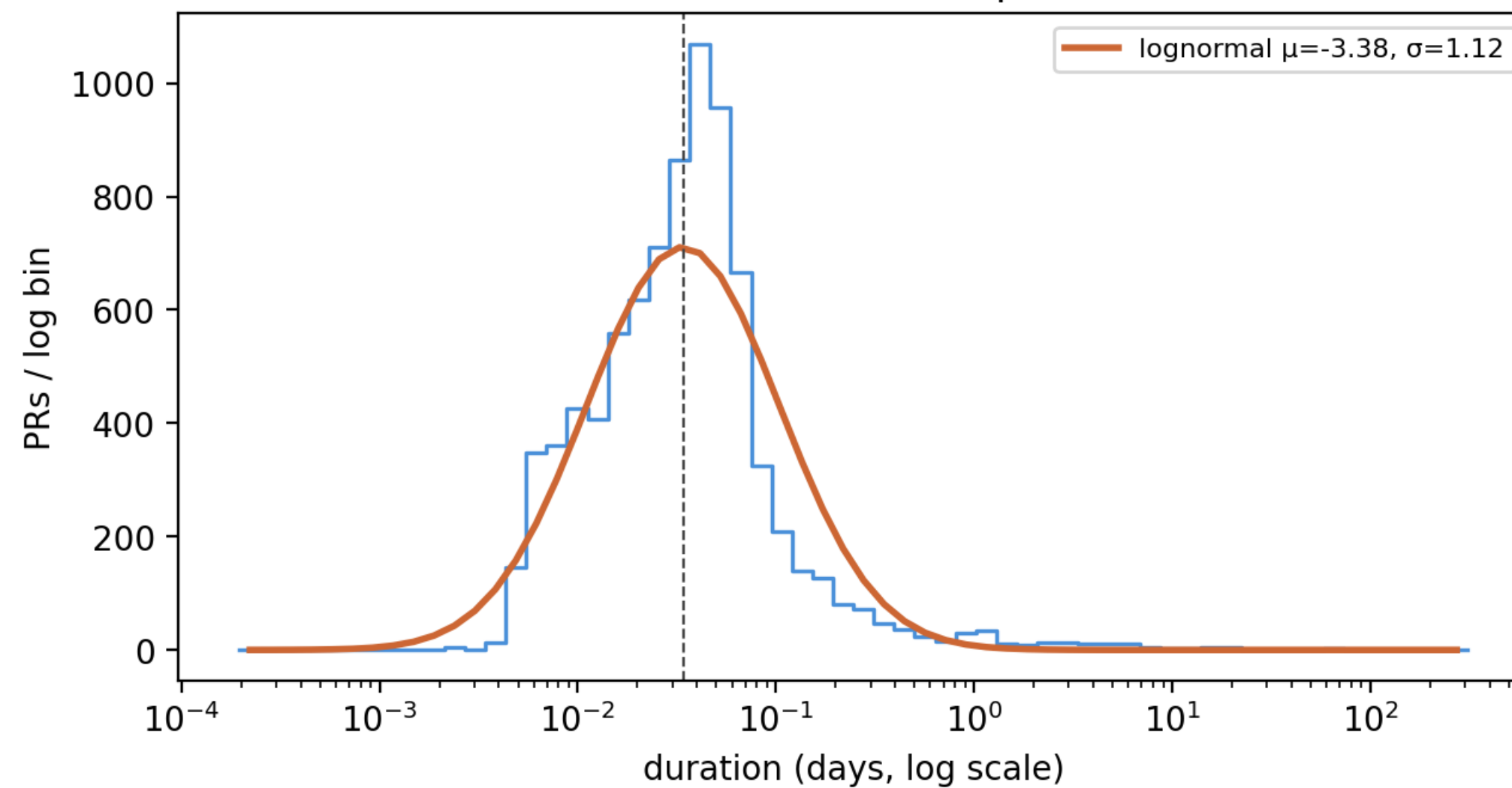
RTM → merged
n=18542, median=0.03d, p90=0.10d



ready-to-merge -> merged

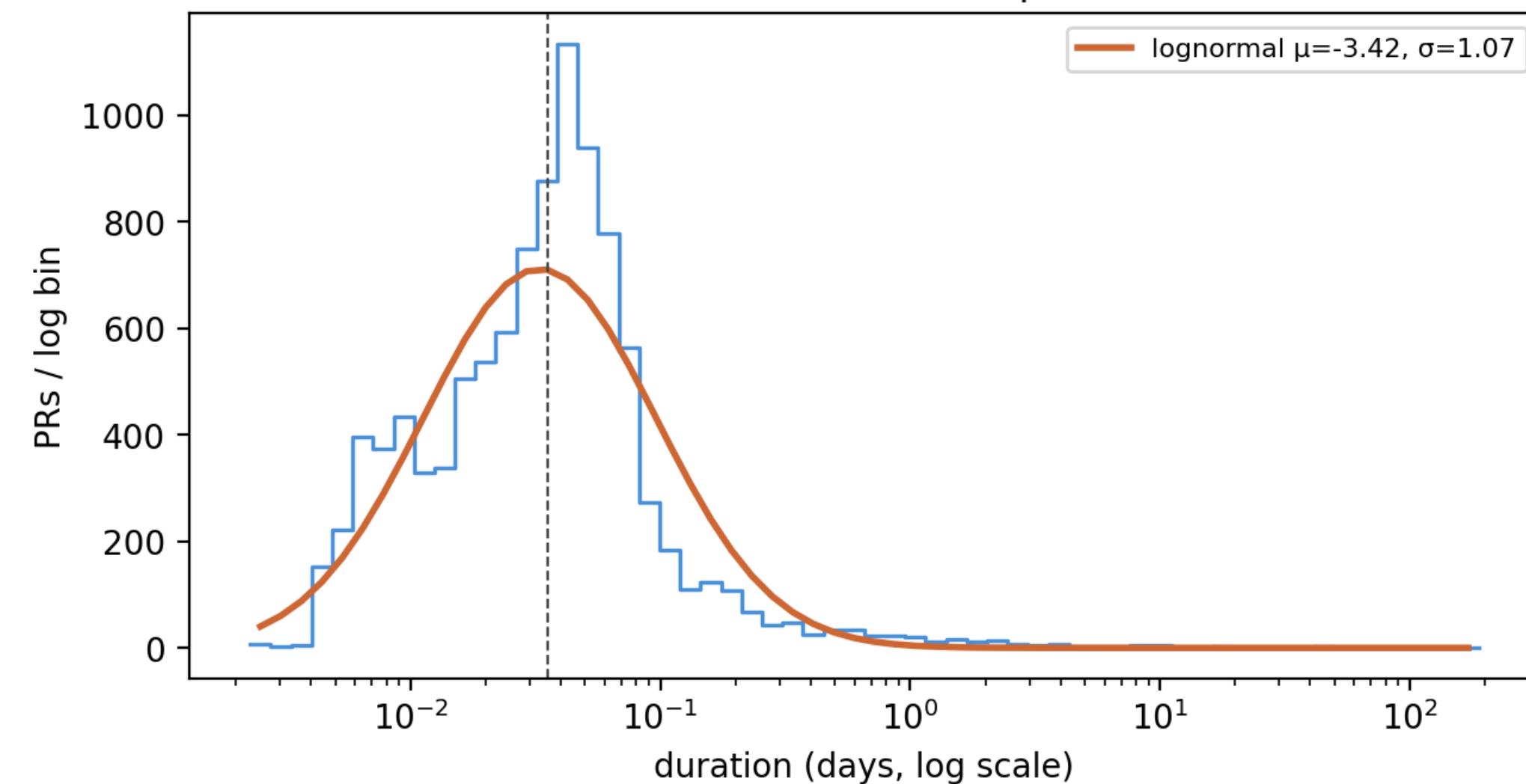
feat (49 mins)

RTM → merged
n=8395, median=0.03d, p90=0.10d



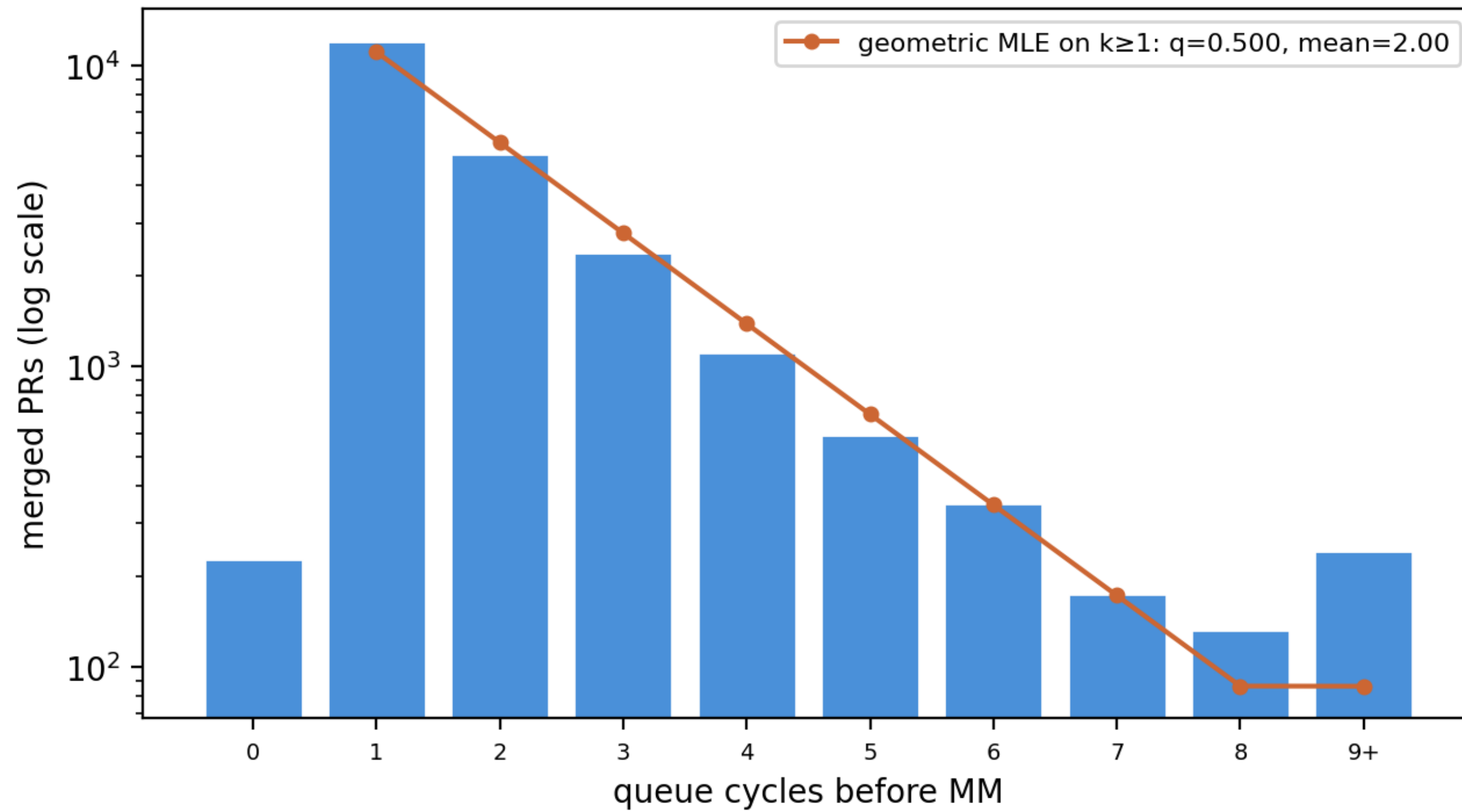
non-feat (51 mins)

RTM → merged
n=10147, median=0.04d, p90=0.09d

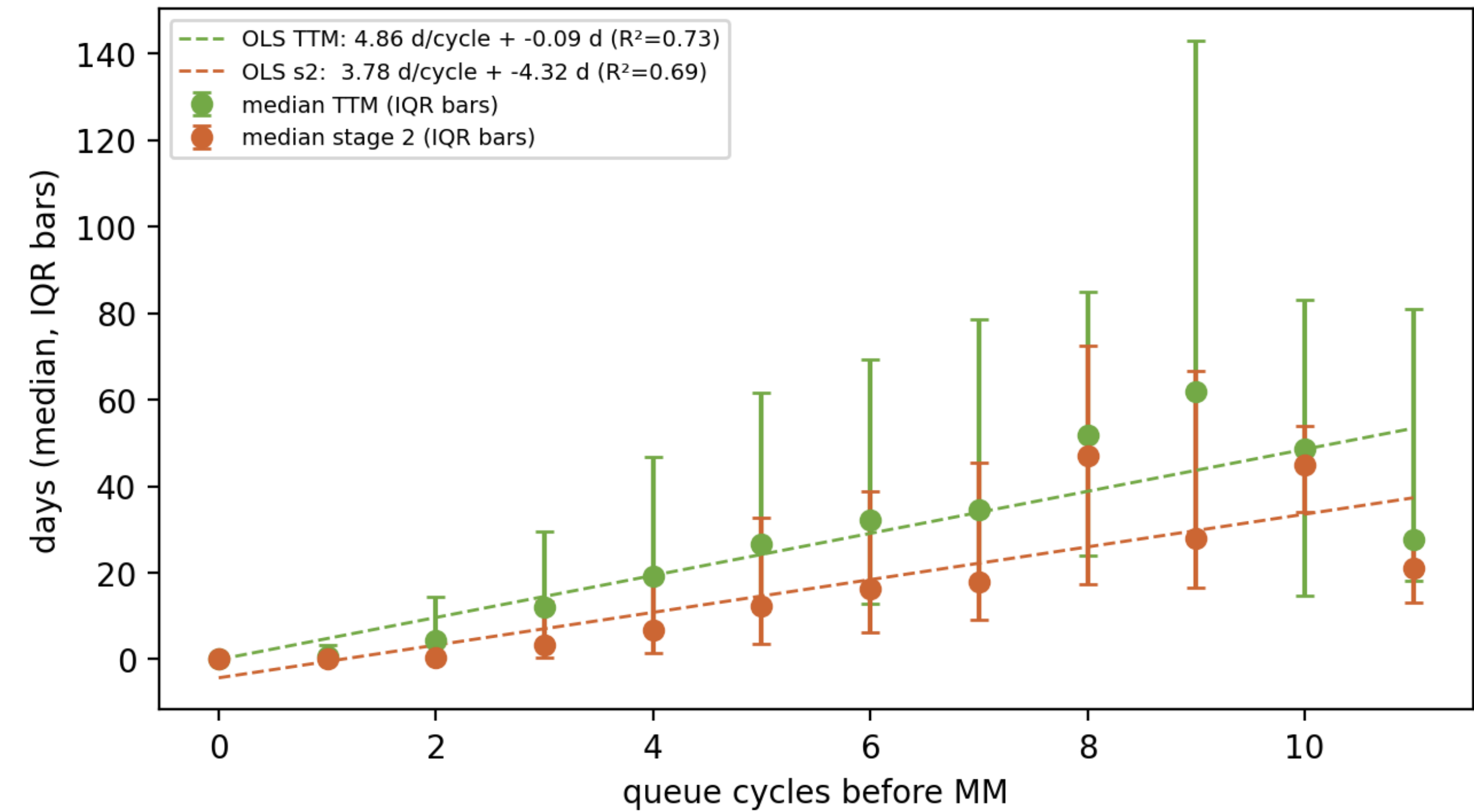


Thank you!

Cycle count distribution (merged, log Y)



TTM & stage 2 vs cycle count (buckets with $n \geq 30$, OLS fit)



Pre-MM queue-window duration by cycle index

